

Emerging anchored FRP systems bonded to steel subjected to monotonic and cyclic loading: A numerical study

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ABSTRACT

Recently, the study of the bond behaviour of hybrid joints has increased due to their application in different industries. Their main purpose is to obtain lightweight but strong, durable structures. In addition, in some industries such as the automotive industry, those requirements may facilitate the construction of vehicles that have lower carbon dioxide emissions. Although the knowledge on the bond behaviour of hybrid bonded joints under monotonic loading is sufficient, the knowledge on the cyclic bond behaviour needs to be improved. The present work aims to mitigate that gap by proposing a numerical model in which the cyclic bond performance between a Carbon Fiber Reinforced Polymer (CFRP) bonded to a steel substrate can be analysed. The results provided by the Externally Bonded Reinforcement (EBR) are used as reference data. These other simulated bonding techniques cover cases where the CFRP is anchored to the substrate in different ways, which are becoming more popular, namely by: (i) linearly increasing the width of the CFRP; (ii) using the mixed adhesive concept (two solutions were considered); (iii) using a steel plate on top of the CFRP strip; and (iv) assuming no interfacial slips at the CFRP unpulled end, which is intended to simulate a perfect anchorage. Compared to the simulations carried out under monotonic loading, the simulations with the adopted cyclic loading history (loading/unloading cycles), allowed us to observe a degradation of the bond strength of the joints with the number of cycles. However, if the overlapped bonded joint is long enough, the strength of the CFRP-to-steel joints is not affected. Excluding the numerical specimens with the perfectly anchored CFRP, i.e. with an “ideal” anchorage, the length of the other adopted anchorages affected either the strength or the ductility of the joint, whether subjected to a monotonic or to the adopted cyclic loading protocol.

1. Introduction

The first uses of adhesives go back to 4,000 BCE [1] but only in the last century its application on different industries grew significantly. Nowadays, with the development of new lightweight and durable materials such as Carbon Fiber Reinforced Polymers (CFRP), the bonding technique allow us to build high-strength lightweight multi material structures free of stress concentration issues for the automotive, construction, aerospace or naval industries.

In the literature [2–10], several joining configurations with different geometries, materials, layers, etc. can be found. Usually, these

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adhesively bonded joints are made by bonding two different materials (or adherends) with an adhesive over a bonded area, which is usually referred to as a hybrid joint in the literature [11–13]. Although the strength of each configuration can be easily tested by monotonic loading, a complete understanding of the bond performance has not been acquired. Furthermore, since different loading types are expected to act on adhesively bonded structures during their lifetime, the design conditions of these joints are of utmost importance, and for this reason, some authors have already made some analytical [14–17] or numerical [18–22] proposals to better understand the bond behaviour of hybrid bonded joints when subjected to a mechanical load [23–28], fatigue [29–35], temperature variation [36–39], creep [40–42] or natural ageing within a controlled laboratory environment [43–47].

Although some studies on CFRP to concrete bonded joints subjected to cyclic loading can be found in the literature, e.g. [33–35,48–50], less information is available in the literature about the cyclic bond behaviour of CFRP-to-steel joints [51]. This is important because the main failure mechanism of CFRP-to-concrete bonded joints (cohesive failure in a thinly layered surface of concrete) is different from the CFRP-to-steel bonded joints (cohesive failure in the adhesive). Some relevant studies covering hybrid bonded joints between a CFRP composite and a steel plate or bar can be found in the references [51–59].

For instance, Doroudi et al. [57] carried out a series of tests where the CFRP-to-steel bonded joints were subjected to a fatigue load. The authors [57] concluded that the maximum bond stress developed within the bonded interface decreases with the fatigue load. Moreover, the experimental results also revealed that the interfacial bond stresses may develop a negative value upon unloading the specimens to a zero load situation. This indicates that the interfacial bond stresses can be reversed during the unloading path of cyclic loading even when the interfacial slips stay positive.

Yang et al. [54,55] also carried out a series of experiments with CFRP-to-steel bonded joints subjected to two types of cyclic loadings. The load capacity of the CFRP-to-steel joints hardly changed with the cyclic loadings which was justified by the long bonded length used, i.e. longer than the effective bond length. Beyond an elastic limit, the global (load-slip) and local (bond-slip) behaviours showed a stiffness loss with the number of cycles imposed to the specimens.

From the point of view of the numerical simulations, the strategies followed by researchers [34,35,48,57] are all based on a damage model where plastic interfacial slips are considered in the local bond-slip relationship. However, the main issue with those numerical simulations lies in the fact that the models were originally developed for CFRP-to-concrete joints rather than CFRP-to-steel joints [57]. Since the failure modes in these two adhesively bonded structures (concrete and steel) are different, the characteristics of the local bond-slip relationship should be different as well. Supported by the experimental results, Doroudi et al. [57] have proposed a local bond-slip formulation that includes the development of reversed interfacial bond stresses, which allows the model to estimate negative stresses within the bonded interface during the application of a fatigue load.

However, the way that the unloading/reloading paths are locally simulated is not yet consensual. For instance, a different perspective was proposed by Carrara and De Lorenzis [34], where their Cohesive Zone Model (CZM) is defined through a bond-slip model that does not assume the development of negative bond stresses. In this case, the authors [34] developed a bond-slip relationship in which negative bond stresses are restricted. For instance, during an unloading path, if the bond stresses tend to go into the negative region under positive slips, then the interfacial slips will decrease under zero bond stress. According to Carrara and De Lorenzis [34], this is a model that provides a solution that agrees well with the experimental findings obtained by Ko and Sato [29], who have studied the bond performance of FRP sheets externally bonded to concrete. It is noteworthy that unlike the proposed bond-slip model of Doroudi et al. [57], the proposed model of Carrara and De Lorenzis [34] shows hysteresis loops during the unloading/reloading paths of the cyclic loading. This is an important aspect that should be considered not only because it leads to a degradation of the loads transmitted to the hybrid bonded joints but also because it intends to better represent the reality.

From the aforementioned numerical formulations, an interesting aspect that researchers seem to agree on is the definition of a function that can accurately correlate the local damage with the local dissipated energy. When adjusted to the experiments, the local damage vs. dissipated energy seems to follow a 2nd-order polynomial function [34,35], which, applied to the simulation of the debonding process between a CFRP composite and a steel substrate, can predict the global load-slip behaviour of these bonded interfaces fairly well. This is one route that researchers may follow to numerically extrapolate the numerical simulations to other situations and increase, therefore, the current knowledge of cyclic bond behaviour of CFRP-to-steel joints subjected to different load conditions. Along these lines, the model proposed by Martinelli and Caggiano [33] assumes a local bond-slip relationship in which the degradation of the interfacial stiffness during the unloading phase depends also on the value of the dissipated energy in each point of the interface.

Previous studies about CFRP-to-steel bonded joints demonstrated the reinforcement benefit provided by the composite and that cyclic loading affects the behaviour of bonded interfaces. Damage plasticity and negative stresses during reversal need to be considered in predictive analysis. The present work aims, therefore, to propose a numerical strategy that allow us to better understand the cyclic bond behaviour of different bonding emerging systems in which the FRP unpulled end is anchored to the steel structure. To that end, an alternative bond-slip model to those already mentioned is used. The simulations are based on the CZM, in which the local bond-slip relationship combines both particularities already mentioned for the models of Carrara and De Lorenzis [34] and Doroudi et al. [57], i.e. it can deal with the development of negative bond stresses within the bonded interface with positive interfacial slips, and, at the same time, it also assumes a hysteresis loop during the unloading/reloading paths of the loading cycle. A numerical programme with a total of 6 different bonding techniques contemplating 48 different models was defined. To evaluate the efficiency of each bonding technique, the first series of specimens corresponds to the specimens where the Externally Bonded Reinforcement (EBR) was simulated and their results are used as the reference. Moreover, monotonic and cyclic loading were considered, which allowed us to analyse the influence of the cyclic bond behaviour on the specimens by comparing their results with the homologous specimens, i.e. with those specimens with the same total bonded length subjected to a monotonic loading until failure.

With all simulated emerging bonding techniques, the results herein reported are therefore important to help us to understand the

debonding process of all six different types of specimens with different bonding typologies. This work also allows us to identify what the most promising bonding technique to be used on steel structures is by revealing the advantages and disadvantages raised by the numerical data in each series of specimens. Therefore, the analysis performed in this work is mainly based on the global load-slip response, degradation of the global stiffness of the joints, the local and global damage to the CFRP-to-steel joints, and dissipated energy.

2. Implemented numerical strategy

2.1. Key assumptions

A theoretical model is proposed to simulate the cyclic bond behaviour between an FRP composite and a substrate. In the present case, the model is developed with the main following assumptions:

- (i) when compared with the FRP composite, the axial stiffness of the substrate is very high and can be assumed to be infinite, i.e. a rigid substrate is considered;
- (ii) the interfacial cracking between both materials develops within a pure shear mode usually denoted as pure mode II;
- (iii) the thickness of both materials undergoes no change during the debonding process and the distortion of the FRP composite is ignored;
- (iv) the interfacial bond stresses developed along the width of the FRP composite, i.e. perpendicularly to the direction of the bonded length, remain constant;
- (v) the monotonic bond behaviour of the interface is defined using an analytical model with an initial elastic stage followed by a linear softened branch;
- (vi) each contact in the interface is cohesively modelled using the bilinear stress vs. displacement model described in the previous item which makes it possible to create hybrid bonded joints with multiple local bond behaviours;
- (vii) the degradation of the bonded joint depends on the accumulated plastic displacement developed in each point of the interface.

All the aforementioned key assumptions are usually used to define the governing equation of a hybrid bonded joint such as CFRP-to-steel joints. To define that equation, classical equilibrium and compatibility conditions are used and the FRP composite is assumed to have a linear stress-strain relationship with a fragile rupture.

2.2. The debonding governing equations

As mentioned before, the proposed numerical model aims to simulate the cyclic bond behaviour of hybrid bonded joints such as CFRP-to-steel joints where the CFRP is bonded to the steel using different bonding techniques. To obtain a general and versatile solution, the width of the FRP composite is assumed to have a linear variation along its length as illustrated in Fig. 1. It should be noted also that by assuming a smooth linear variation this bonding solution may avoid also the development of undesired shear stresses along the CFRP length that may lead to the shear rupture of the CFRP composite [60]. Given this assumption, both situations can be modelled, i.e. assuming a bonded length where the FRP composite has a constant and linear variable width.

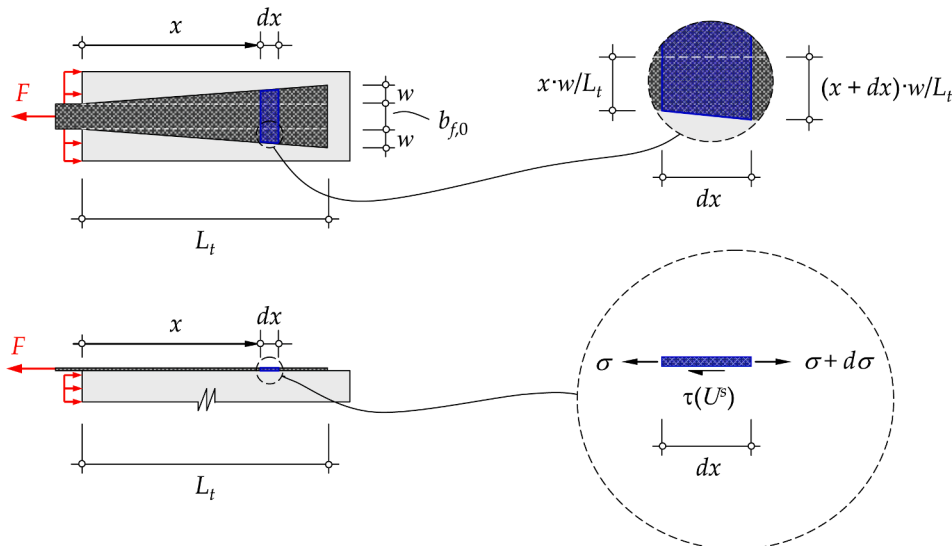


Fig. 1. Hybrid bonded joint with variable width along the bonded length.

Before defining a general governing equation of the debonding between the FRP composite and the steel substrate, let us consider first an initial width of the FRP composite $b_{f,0}$ with a width increment rate of w/L_t . Thus, the width of the CFRP at its unpulled end is $b_{f,0} + 2w$. Considering the finite segment dx illustrated in Fig. 1, the equilibrium of the forces in the horizontal direction (i.e. in the x -axis direction) leads to the following equilibrium equation:

$$\sigma \cdot t_f \cdot b_f(x) + \tau(U^s) \cdot \frac{dx}{2} \cdot (b_f(x) + b_f(x + dx)) - (\sigma + d\sigma) \cdot t_f \cdot b_f(x + dx) = 0 \quad (1)$$

where $\tau(U^s)$ is the interfacial bond stress developed in the finite segment dx , which is dependent on the local bond-slip relationship of the bonded joint; and $b_f(x)$ and $b_f(x + dx)$ represent the widths of the FRP composite at points x and $x + dx$, respectively defined as follows:

$$b_f(x) = b_{f,0} + 2 \cdot x \cdot \frac{w}{L_t} \quad (2)$$

and

$$b_f(x + dx) = b_{f,0} + 2 \cdot (x + dx) \cdot \frac{w}{L_t} \quad (3)$$

Introducing Eqs. (2) and (3) into Eq. (1) and ignoring all the second-order infinitesimals in Eq. (1), the following equation is obtained:

$$\sigma_{,x} + \frac{2\sigma \cdot \frac{w}{L_t}}{b_{f,0} + 2x \cdot \frac{w}{L_t}} - \frac{\tau(U^s)}{t_f} = 0 \quad (4)$$

where $\sigma_{,x}$ is the first derivative of the axial stresses developed in the cross-section of the FRP composite. The interfacial slip developed within the bonded interface is obtained by the relative displacement between materials [61-65], which differentiate with respect to x and, as already mentioned, so assuming that the substrate is rigid, the following expression can be established:

$$s_{,x} = u_{f,x} \quad (5)$$

where $s_{,x}$ is the first derivative of the relative displacements (or interfacial slips) between adherends with respect to x ; and $u_{f,x}$ represents the strains in the FRP composite. Since FRP composites have a linear stress-strain relationship up until failure, the axial stresses developed in the FRP are:

$$\sigma = E_f \cdot u_{f,x} = E_f \cdot s_{,x} \quad (6)$$

Differentiating Eq. (6) with respect to x and introducing it into Eq. (4), the following governing equation of the system is finally obtained:

$$s_{,xx} + \frac{1}{C + x} \cdot s_{,x} - \frac{\tau(U^s)}{E_f \cdot t_f} = 0 \quad (7)$$

where $s_{,xx}$ is the second derivative of the interfacial slips with respect to x ; and C is given by:

$$C = \frac{b_{f,0} \cdot L_t}{2 \cdot w} \quad (8)$$

It is worth mentioning that if the particular case of $w = 0$ mm is considered, i.e. if the width of the FRP composite is constant throughout the total bonded length of the overlapped bonded joint, then the following well-known governing equation is obtained, e.g. [51,66-69]:

$$s_{,xx} - \frac{\tau(U^s)}{E_f \cdot t_f} = 0 \quad (9)$$

The use of the described model has two other important advantages. The first is that it allows us to simulate the monotonic and cyclic bond behaviour of a bonded solution where the FRP composite may have a constant and variable width. Such a combined solution consists of one bonded joint where the width of the FRP composite is constant in the first part of the bonded length ($L_{b,c}$) whilst the width of the FRP composite varies linearly ($L_{b,v}$) in the second part, i.e. between $L_{b,c}$ and $L_{b,v}$. Given this hypothesis, the debonding process of these bonded joints is governed, therefore, by combining Eqs. (7) and (9). In addition, the use of a mixed adhesive or an additional rigid layer is another possibility that can be considered if, within those same bonded length limits, two different local bond-slip models are assumed and/or if an additional stiff layer is placed on the top of the FRP composite.

2.3. Proposed cohesive zone modelling

The modelling of the bond between the CFRP composite and the substrate is based on Cohesive Zone Models (CZM) with a bilinear shape, i.e. the relationship between the bond stresses developed at a certain point of the interface and the slips at that same point.

Under a monotonic load, the interfacial slips developed at each point of the overlapped bonded joint should follow, therefore, the bond-slip relationship shown in Fig. 2a. The same bilinear bond-slip model was used to represent the cyclic bond behaviour between a CFRP composite and a rigid substrate, such as a steel frame, with an unloading–reloading path parallel to the linear elastic branch. As illustrated in Fig. 2a, upon monotonic loading, the elastic branch stiffness of the proposed model is k_e^s and after reaching $\tau_{\max}$, it decreases linearly to $\tau = 0$ following a linear decrease with a softening stiffness herein denoted as k_d^s . The interfacial slip at $\tau_{\max}$ is denoted as $U_{\max}^s$, whereas at failure is U_{ult}^s . Upon cyclic loading, shown in Fig. 2b, with repeated unloading and reloading, the unloading–reloading stiffness is k_e^s . However, it is assumed that the stress threshold value ($\tau_{\max,n}$) decreases with the accumulated plastic displacement ($U_{p,n}^s$) of the interface at cycle n according to a linear degradation of the maximum bond with respect to the softening interval ($U_{ult}^s - U_{\max}^s$), i.e.:

$$\frac{\tau_{\max}(U_{p,n}^s)}{\tau_{\max}} = \pm \left(1 - \frac{U_{p,n}^s}{U_{ult}^s - U_{\max}^s} \right) \quad (10)$$

The plastic displacement accumulation initiates first when the maximum shear stress is attained at $U_{\max}^s$, and then increases in subsequent reloading or unloading paths if the updated stress threshold value is reached. The maximum accumulated plastic displacement corresponds to the amplitude of the softening branch of the bilinear bond-slip model, i.e. $U_{ult}^s - U_{\max}^s$, in which case the interface will fail at that point. Furthermore, to better illustrate the CZM upon cyclic loading, Fig. 2c shows the evolution of the model variables with respect to time. Each branch of bond stress and interfacial slip (total and plastic) corresponds to a particular part of the bond-slip model shown in Fig. 2b. The upper and lower bond threshold curves follow Eq. (10).

For instance, for the n th cycle, at the end of the (re)loading path (simulation point $n - 0.5$) with the slip U_n^s the interfacial bond stress at any point of the interface is:

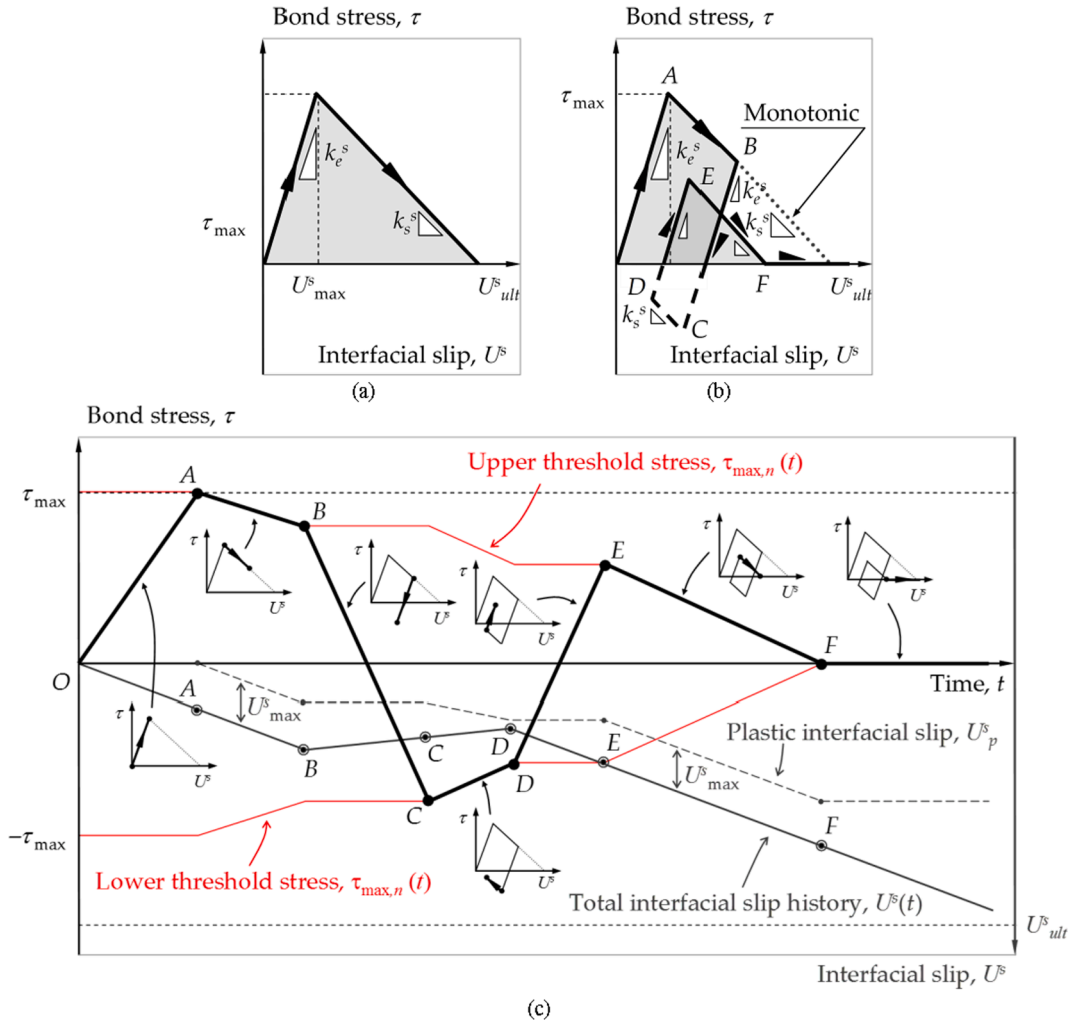


Fig. 2. The bilinear CZM: (a) subjected to monotonic loading; (b) subjected to a cyclic loading example; and (c) evolution with respect to time when subjected to a cyclic loading example.

$$\tau(U^s) = \begin{cases} k_e^s U_{p,n}^s & \text{if } U_{p,n}^s = 0 \\ \tau_{n-0.5} + k_e^s \cdot (U_n^s - U_{n-0.5}^s) & \text{if } U_{p,n}^s = U_{p,n-0.5}^s \\ \tau_{max,n} - \tau_{max} \cdot \left(\frac{U_{p,n}^s - U_{p,n-0.5}^s}{U_{ult}^s - U_{max}^s} \right) & \text{if } U_{p,n-0.5}^s < U_{p,n}^s < U_{ult}^s - U_{max}^s \end{cases} \quad (11)$$

and the accumulated plastic displacement is:

$$U_p^s = U_{p,n-0.5}^s + \begin{cases} 0 & \text{if } U_n^s < U_{n-0.5}^s + \frac{\tau_{max,n} - \tau_{n-0.5}}{k_e^s} \\ U_n^s - \left(U_{n-0.5}^s + \frac{\tau_{max,n} - \tau_{n-0.5}}{k_e^s} \right) & \text{if } U_n^s > U_{n-0.5}^s + \frac{\tau_{max,n} - \tau_{n-0.5}}{k_e^s} \end{cases} \quad (12)$$

while at the end of the unloading path (simulation of the n th loading cycle) the interfacial bond stress at any of the points within the interface is:

$$\tau(U^s) = \begin{cases} k_e^s U_{p,n}^s & \text{if } U_{p,n}^s = 0 \\ \tau_{n-0.5} - k_e^s \cdot (U_{n-0.5}^s - U_n^s) & \text{if } U_{p,n}^s = U_{p,n-0.5}^s \\ \tau_{max,n} + \tau_{max} \cdot \left(\frac{U_{p,n}^s - U_{p,n-0.5}^s}{U_{ult}^s - U_{max}^s} \right) & \text{if } U_{p,n-0.5}^s < U_{p,n}^s < U_{ult}^s - U_{max}^s \end{cases} \quad (13)$$

and the corresponding accumulated plastic displacement is:

$$U_p^s = U_{p,n-0.5}^s + \begin{cases} 0 & \text{if } U_n^s \geq U_{n-0.5}^s + \frac{\tau_{max,n} - \tau_{n-0.5}}{k_e^s} \\ -U_n^s + \left(U_{n-0.5}^s + \frac{\tau_{max,n} - \tau_{n-0.5}}{k_e^s} \right) & \text{if } U_n^s < U_{n-0.5}^s + \frac{\tau_{max,n} - \tau_{n-0.5}}{k_e^s} \end{cases} \quad (14)$$

2.4. Definition of the damage

The damage condition of the adhesively anchored CFRP systems is monitored on two different scales of analysis: locally at every CFRP to substrate (discrete) contact in the numerical model and globally for the entire structural system. From the description of the contact law adopted to model the interface between the CFRP and the substrate, discrete interfacial damage is null when (re)loading-unloading is always within the elastic branch and increases proportionally to the accumulated plastic displacement when the maximum shear stress is attained. Hence, using the same damage quantification method used in the literature [70], the local damage condition of every CFRP to substrate contact varies between 0 and 1, and at cycle n (D_n^U) is:

$$D_n^U = \frac{U_{p,n}^s}{U_{ult}^s - U_{max}^s} \quad (15)$$

The global damage condition of the adhesively anchored CFRP systems can be evaluated from the damage condition of each component separately (CFRP and substrate), and the adhesively bonded interface. Regarding the individual components, two different situations are identified. The substrate was considered to be rigid, and damage could not develop within this component. The CFRP behaviour under tension was modelled (see Fig. 1a) with a linear elastic model with fragile rupture, which means that damage is null up to the point of failure, and total (equal to 1) when the FRP reaches its rupture. The global damage condition of the interface was determined by integrating Eq. (15) for the bonded area. Then, dividing it by the total bonded area, the global damage ($D_{g,n}$) developed in the specimens in each loading cycle n is determined by the following equation:

$$D_{g,n} = \frac{\int D_n^U dA}{\int dA} \quad (16)$$

where D_n^U is determined from Eq. (15) and dA is an infinitesimal bonded area of the overlapped bonded joint.

2.5. Numerical procedure

Since the implementation of the bond-slip relationship shown in Fig. 2b into Eqs. (7) and/or (9) requires the use of a numerical strategy, the discrete element method was selected to study the effect of different bonding techniques on the monotonic and cyclic bond behaviour of CFRP-to-steel joints. This method was originally introduced by Cundall in the 1970 s for the study of rock fracture problems, and since then it has been adapted for the study of different other engineering problems [71]. It consists of representing the materials by an assembly of rigid particles which interact with each other through soft contact points, and the displacement and rotation of each particle are related to the set of forces acting on it. This means that if the assembly is in static equilibrium there is no motion.

Based on Newton's second law, it applies to each particle the equation of translational motion [72],

$$\mathbf{F}_{i,t} = m \cdot \ddot{\mathbf{x}}_{i,t} \quad (17)$$

where $\mathbf{F}_{i,t}$ is the total applied force vector in i co-ordinate directions (1, 2) at time t ; m is the mass of the particle; and $\ddot{\mathbf{x}}_{i,t}$ is the linear acceleration vector of the particle. It is worth mentioning that in the present work the particles have no rotation and, therefore, the corresponding equation of rotational motion

$$\mathbf{M}_{3,t} = I \cdot \ddot{\omega}_{3,t} \quad (18)$$

was not implemented, in which $\mathbf{M}_{3,t}$ is the moment in the co-ordinate direction 3 at time t ; I is the moment of inertia of particle concerning its gravity centre; and $\ddot{\omega}_{3,t}$ represents the angular acceleration vector of the particle.

The equation of translational motion in (17) is solved using the central-difference method to determine the accelerations, velocities, and displacements of the particles. Hence, the linear accelerations of the particles at time t are a function of the velocities vector ($\dot{\mathbf{x}}_i$) and the time step (Δt), as:

$$\ddot{\mathbf{x}}_{i,t} = \frac{1}{\Delta t} \cdot (\dot{\mathbf{x}}_{i,t+\Delta t/2} - \dot{\mathbf{x}}_{i,t-\Delta t/2}) \quad (19)$$

To obtain the linear velocities of the particles at time $t + \Delta t/2$, Eq. (19) is introduced into Eq. (17) which yields:

$$\dot{\mathbf{x}}_{i,t+\Delta t/2} = \dot{\mathbf{x}}_{i,t-\Delta t/2} + \frac{\mathbf{F}_{i,t}}{m} \Delta t \quad (20)$$

Therefore, the updated position of a particle is determined according to:

$$\mathbf{x}_{i,t+\Delta t} = \mathbf{x}_{i,t} + \dot{\mathbf{x}}_{i,t+\Delta t/2} \cdot \Delta t \quad (21)$$

The displacements at the particle's point of contact are then determined. It should be kept in mind that since the rotations of the particles are ignored, their angular velocities and rotation angles are not calculated during the determination of the updated positions of the particles. In addition, by considering circular particles, the formulation is significantly simplified [71].

The contact between particles A and B is illustrated in Fig. 3 with greater detail on the right-hand side, and the contact position vector, $\mathbf{x}_{i,t}^{[C]}$, at time t is defined as [73]:

$$\mathbf{x}_{i,t}^{[C]} = \mathbf{x}_{i,t}^{[A]} + \left(R^{[A]} - \frac{1}{2} \cdot U_t^n \right) \cdot \mathbf{n}_i \quad (22)$$

where the contact overlap in the normal direction, U_t^n , is:

$$U_t^n = R^{[A]} + R^{[B]} - d \quad (23)$$

and $\mathbf{n}_i$ is the contact unit normal vector, defined as:

$$\mathbf{n}_i = \frac{\mathbf{x}_{i,t}^{[B]} - \mathbf{x}_{i,t}^{[A]}}{d} \quad (24)$$

and $\mathbf{x}_i^{[A]}$ and $\mathbf{x}_i^{[B]}$ are the centre position vectors in coordinates i of particles A and B, respectively; $R^{[A]}$ and $R^{[B]}$ are the radius of particles A and B, respectively; and d is the distance between the centres of particles A and B. To calculate the velocity of the contact at time $t + \Delta t/2$ the following equation can be used [73]:

$$\dot{\mathbf{x}}_{i,t+\Delta t/2}^{[C]} = \dot{\mathbf{x}}_{i,t+\Delta t/2}^{[B]} - \dot{\mathbf{x}}_{i,t+\Delta t/2}^{[A]} \quad (25)$$

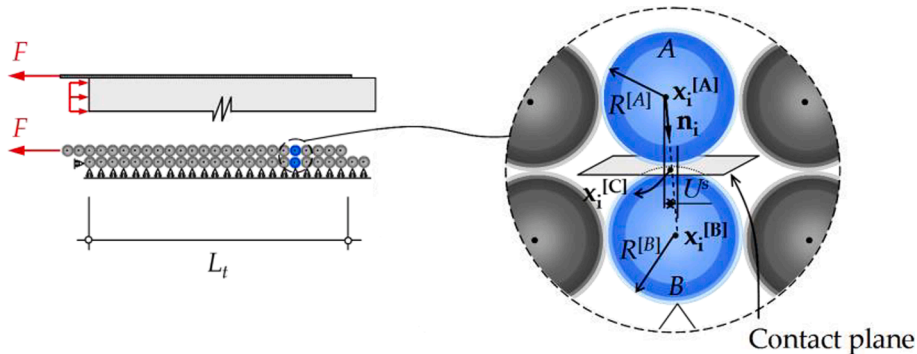


Fig. 3. Discrete Element Method for the bond between the FRP composite and the steel substrate.

Table 1

Identification of the numerical models.

Series	ID of the numerical model	Type of loading	Total bonded length L_t (mm)	Free bonded length L_b (mm)	Anchorage length L_a (mm)	Modelled bonding technique
1	REF-CFRP/Steel-L50:M	Monotonic	50	50	–	EBR
	REF-CFRP/Steel-L100:M		100	100	–	
	REF-CFRP/Steel-L200:M		200	200	–	
	REF-CFRP/Steel-L250:M		250	250	–	
	REF-CFRP/Steel-L300:M		300	300	–	
	REF-CFRP/Steel-L400:M		400	400	–	
	REF-CFRP/Steel-L50:C	Cyclic	50	50	–	
	REF-CFRP/Steel-L100:C		100	100	–	
	REF-CFRP/Steel-L200:C		200	200	–	
	REF-CFRP/Steel-L250:C		250	250	–	
2	REF-CFRP/Steel-L300:C		300	300	–	
	REF-CFRP/Steel-L400:C		400	400	–	
	VW-CFRP/Steel-L50/50:M	Monotonic	100	50	50	EBR with an anchored CFRP strip with Variable Width (VW)
	VW-CFRP/Steel-L250/50:M		300	250	50	
	VW-CFRP/Steel-L50/150:M		200	50	150	
	VW-CFRP/Steel-L250/150:M		400	250	150	
	VW-CFRP/Steel-L50/50:C	Cyclic	100	50	50	
	VW-CFRP/Steel-L250/50:C		300	250	50	
	VW-CFRP/Steel-L50/150:C		200	50	150	
	VW-CFRP/Steel-L250/150:C		400	250	150	
	MA(s1)-CFRP/Steel-L50/50:M	Monotonic	100	50	50	EBR with a Mixed Adhesive (MA) - solution 1 (s1)
	MA(s1)-CFRP/Steel-L50/150:M		300	250	50	
	MA(s1)-CFRP/Steel-L250/50:M		200	50	150	
	MA(s1)-CFRP/Steel-L250/150:M		400	250	150	
	MA(s1)-CFRP/Steel-L50/50:C	Cyclic	100	50	50	
	MA(s1)-CFRP/Steel-L50/150:C		300	250	50	
	MA(s1)-CFRP/Steel-L250/50:C		200	50	150	
	MA(s1)-CFRP/Steel-L250/150:C		400	250	150	
4	MA(s2)-CFRP/Steel-L50/50:M	Monotonic	100	50	50	EBR with a Mixed Adhesive (MA) - solution 2 (s2)
	MA(s2)-CFRP/Steel-L50/150:M		300	250	50	
	MA(s2)-CFRP/Steel-L250/50:M		200	50	150	
	MA(s2)-CFRP/Steel-L250/150:M		400	250	150	
	MA(s2)-CFRP/Steel-L50/50:C	Cyclic	100	50	50	
	MA(s2)-CFRP/Steel-L50/150:C		300	250	50	
	MA(s2)-CFRP/Steel-L250/50:C		200	50	150	
	MA(s2)-CFRP/Steel-L250/150:C		400	250	150	
5	SPA-CFRP/Steel-L50/50:M	Monotonic	100	50	50	EBR with a Steel Plate Anchorage (SPA)
	SPA-CFRP/Steel-L50/150:M		200	50	150	
	SPA-CFRP/Steel-L250/50:M		300	250	50	
	SPA-CFRP/Steel-L250/150:M		400	250	150	
	SPA-CFRP/Steel-L50/50:C	Cyclic	100	50	50	
	SPA-CFRP/Steel-L50/150:C		200	50	150	
	SPA-CFRP/Steel-L250/50:C		300	250	50	
	SPA-CFRP/Steel-L250/150:C		400	250	150	
6	PA-CFRP/Steel-L50:M	Monotonic	50	50	–	EBR with the CFRP strip Perfectly Anchored (PA)
	PA-CFRP/Steel-L250:M		250	250	–	
	PA-CFRP/Steel-L50:C	Cyclic	50	50	–	
	PA-CFRP/Steel-L250:C		250	250	–	

whereas the following equations can be used to determine the contact displacements in the normal and shear directions, respectively:

$$\Delta x_{t+\Delta t}^{n[C]} = \left(\dot{x}_{i,t+\Delta t/2}^{[C]} \cdot \Delta t \right) \cdot n_i \quad (26)$$

and

$$\Delta x_{i,t+\Delta t}^{s[C]} = \left(\dot{x}_{i,t+\Delta t/2}^{[C]} \cdot \Delta t \right) - \Delta x_{t+\Delta t}^{n[C]} \cdot n_i \quad (27)$$

From this point on, the force–displacement law is applied to update the forces using the incremental linear formulation proposed by Azevedo et al. [74], where the normal force increment is:

$$\Delta F_{t+\Delta t}^{n[C]} = -k^n \cdot \Delta x_{t+\Delta t}^{n[C]} \quad (28)$$

while the shear force increment vector is

$$\Delta F_{i,t+\Delta t}^{s[C]} = -k^s \cdot \Delta x_{i,t+\Delta t}^{s[C]} \quad (29)$$

where k^n and k^s are the normal and shear contact stiffnesses. The force increments are added to the contact forces at a previous time (t), and the resultant contact force vector is:

$$F_{i,t+\Delta t}^{[C]} = F_{i,t}^{[C]} \cdot n_i + F_{i,t+\Delta t}^{s[C]} \quad (30)$$

Finally, the contribution of the contact force to the resultant force of the particles is determined as follows:

$$\begin{cases} F_{i,t+\Delta t}^{[A]} = F_{i,t}^{[A]} - F_{i,t+\Delta t}^{[C]} \\ F_{i,t+\Delta t}^{[B]} = F_{i,t}^{[B]} + F_{i,t+\Delta t}^{[C]} \end{cases} \quad (31)$$

The contact stiffness is defined by the contact model. The contact model being used is most often linear elastic but time-dependent (e.g. viscoelastic) and failure-softening models are also adopted [70,75]. It should be noted that with a soft contact approach the rigid particles overlap each other at contacts when in compression, and the compressive normal contact force is not restrained. On the contrary, a shear contact force or a tensile normal contact force is limited to the maximum value defined by the contact bonding model. Despite this methodology being conceptually simple, it usually demands a higher computational effort than those where continuum-based formulation methodologies are used, and this increases with the number of particles in the assembly as well with the reduction of the time step between calculation cycles.

The left-hand side of Fig. 3 shows the assembly of particles adopted for the modelling of the CFRP-to-steel bonded joints under study. The CFRP composite and the substrate were modelled with equally-sized and aligned circular rigid particles with 1.4 mm diameter which corresponds to the thickness of the FRP composite. As suggested in the literature [28], the discretization of CFRP bonded joints lower than 2.5 mm allowed us to preserve the accuracy of the numerical results and dispense further studies on the mesh-sensitivity of the models. The properties of the contact models used to model the materials are described in Section 3.1. This is a methodology that has already been used in the literature with success to analyse the bond performance of hybrid bonded joints [76-78], where the different mechanical behaviours of the adherends, loading conditions, or the bond-slip models were all considered, which reveals the ability of this method to deal with different situations.

Due to the different bonding techniques considered in this work, different discretizations of the CFRP-to-steel bonded joints were carried out. Therefore, each simulation carried out here was programmed in a specifically developed software [73], which allowed us to analyze the bond performance of different emerging bonding techniques applied to CFRP-to-steel joints subjected either to monotonic or cyclic loading.

3. Numerical programme

3.1. Definition of the adhesively bonded joints

Adhesively bonded joints can be composed of very different materials and adhesion agents, to which several anchorage systems can be applied. Therefore, it was considered important to investigate the mechanical and damage response of different adhesively anchored FRP systems when subjected to cyclic loading. To analyse the bond behaviour between a CFRP and a metallic substrate such as steel, six different situations (or 6 series, as seen in Table 1) were defined. The first one, corresponding to the conventional EBR technique, provides the reference data that will be used for comparison with other cases with different anchorage systems (see Fig. 4a). The second series, which was originally studied by the authors in [60], consists of linearly increasing the width of the CFRP along the overlapped bonded length (see Fig. 4b). It is expected that the specimens in this series may show higher strength than those considered as a reference due to the increase in the bonded area. In the present case, it was assumed that the width of the CFRP composite linearly increases from 10 mm (which is the width used for the CFRP in the reference specimens) to 30 mm over the anchored length. The third and fourth situations are illustrated in Fig. 4c and 4d, respectively, and both correspond to a similar solution in which the concept of mixed adhesive is applied. Thus, part of the overlapped bonded length is bonded with one adhesive, whereas the other part is bonded with another adhesive. The main principle here is to adopt a stiffer adhesive in the opposite region of the CFRP pulled end, which will play the role of the additional anchorage in these two series. In this situation, three different cohesive zone models were used (see Fig. 5). The fifth series simulates the use of a rigid steel plate bonded onto the free surface of the CFRP. The displacements of the right-end point of this rigid steel plate was also constrained (see Fig. 4e). It was considered that no external pressure is applied on the steel plate, which means that the strength increase occurs due to the additional bonded area between the top of the CFRP strip and the steel plate anchorage. Therefore, this series aims to analyse the influence of this additional bonded area on the bond performance of the CFRP-to-steel joint. Finally, the sixth series simulates a theoretical situation where the CFRP unpulled end is perfectly anchored and does not allow the interface to develop interfacial slips between the CFRP strip and the steel substrate at that point (see Fig. 4f). Although this anchorage solution can perfectly constrain the slips at the CFRP unpulled end, from a practical point of view, this is almost impossible to achieve. Nevertheless, and to the best knowledge of the authors, the Continuous Reinforcement Embedded at Ends (CREatE) [79-82] is the closest bonding technique to this theoretical solution introduced in the sixth group of specimens.

To investigate the effect of the bonded length on the mechanical behaviour of adhesively bonded joints, two different bonded lengths were considered for the zones before and within the additional anchorage system. Before the anchorage zone, the bonded lengths were set equal to 50 mm and 250 mm, which represent the situations where the adhesive-bonded joint is lesser or longer than

the effective bonded length (i.e. the length needed to achieve the maximum interfacial load). For the anchorage zones, the bonded length was set equal to 50 mm and 150 mm for the different systems, except for the perfectly anchored system (Series 6) that was nil.

In the numerical simulations, the CFRP behaviour under tension was modelled on previous experimental work reported in the literature [83], where five flat coupons of CFRP laminate from S&P supplier (S&P 150/2000) with a cross-sectional area of 10×1.4 mm (width $\times$ thickness) were tested under a tensile load until rupture. From these tests, the average values of the strain and stress of a CFRP laminate at failure were, respectively, $\varepsilon_{fu} = 1.03\%$ and $\sigma_{fu} = 1,565$ MPa, which results in the calculated average elastic modulus of $E_f = 159$ GPa. The substrate considered in the adhesively bonded joints was a thick steel plate, which means that the value of the CFRP to steel axial stiffness ratio can be considered equal to zero. Hence, the steel plate was numerically modelled as a rigid element.

Based on the thickness of the CFRP laminate adopted in modelling, both the CFRP and the steel plate were numerically modelled with particles with a diameter of 1.4 mm.

To model the adhesive bond behaviour of the interface and anchorage zones, three different CZM with a bilinear shape were used (see Fig. 5). CZM 1, with $\tau_{\max} = 13.27$ MPa, $U^s_{\max} = 0.096$ mm and $U^s_{ult} = 0.256$ mm, is the reference adhesive material, and it defines the behaviour of the interface that is outside the anchorage zone in all models (Series 1 to 6), and the anchorage zone in Series 1, 2 and 5 (EBR, variable width, and steel plate anchorage). Model 2, with $\tau_{\max} = 19.91$ MPa, $U^s_{\max} = 0.144$ mm and $U^s_{ult} = 0.256$ mm, and model 3 with $\tau_{\max} = 19.90$ MPa, $U^s_{\max} = 0.144$ mm and $U^s_{ult} = 0.384$ mm, were both adopted in the anchorage solutions with different adhesive materials. These two latter adhesives have the same elastic stiffness as the reference adhesive but differ in their peak shear stress (50% greater than the reference adhesive), and the ultimate slip in CZM 3 is significantly larger than that in CZM 2. Under these adhesive conditions, it is expected that the bond performance of their corresponding specimens may improve when compared with the reference specimens in Series 1, where the conventional EBR technique was used.

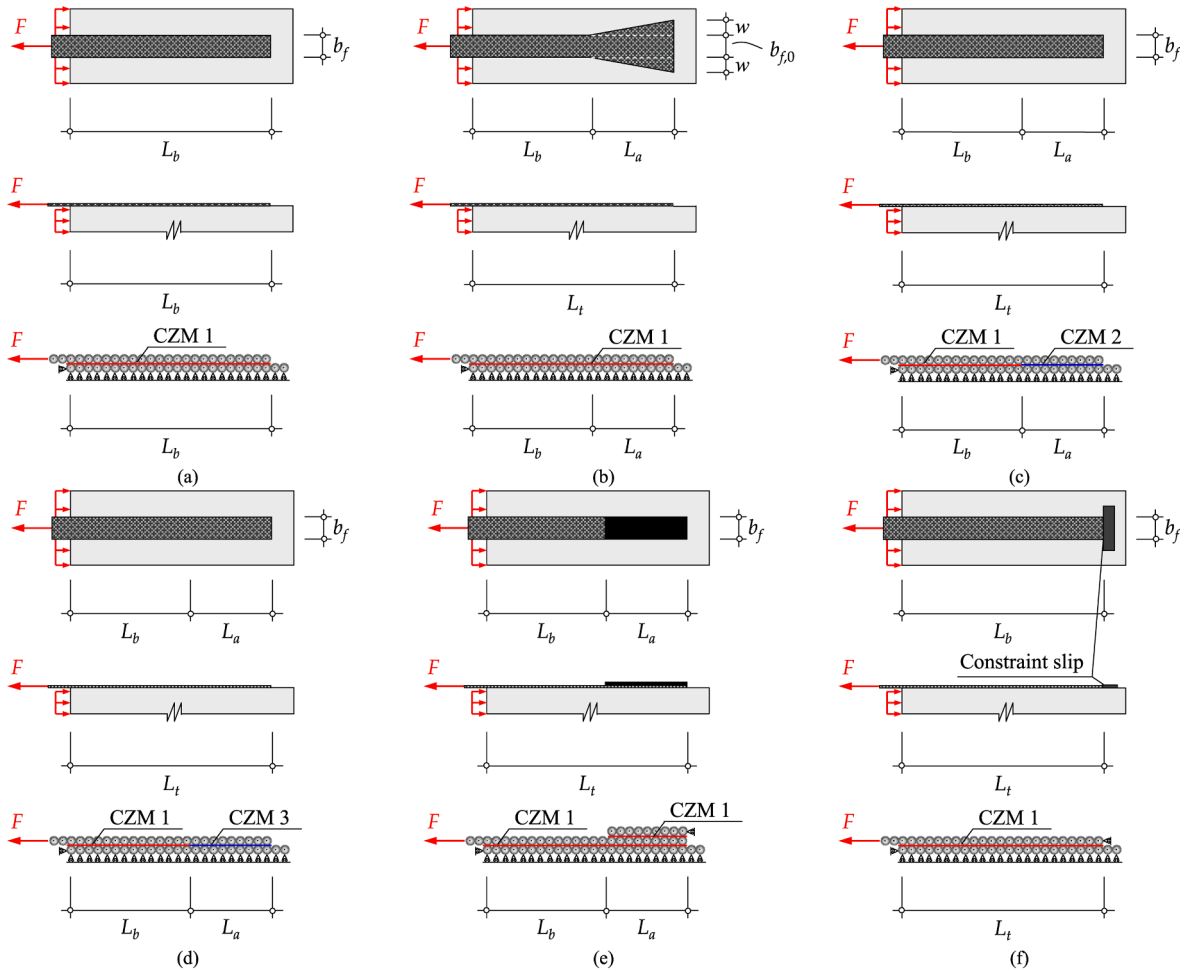


Fig. 4. Modelled specimens in: (a) Series 1; (b) Series 2; (c) Series 3; (d) Series 4; (e) Series 5; and (e) Series 6.

3.2. ID of the numerical simulations

To evaluate the mechanical behaviour of different adhesively anchorage FRP systems the single-lap shear test configuration was adopted. The monotonic and the cyclic loading protocols were considered. Table 1 identifies each of the 48 numerical simulations performed in this work. For an easier analysis and discussion of the results, the numerical simulations are identified in the paper as follows: AJ-CFRP/Steel-Lx:P, where “AJ” corresponds to the adhesive joint system (REF – EBR reference joint solution; VW – EBR with an anchored CFRP strip with Variable Width; MA(s1) – EBR with a Mixed Adhesive – solution 1 (s1); MA(s2) – EBR with a Mixed Adhesive – solution 2 (s2); SPA – EBR with a Steel Plate Anchorage; PA – EBR with the CFRP strip Perfectly Anchored); “CFRP/Steel” correspond to the adherends used in the overlapped bonded joint; “Lx” identifies the bonded length (mm) of the joint and anchorage zones (e.g. EBR – L50, L100, L200, L250, L300, and L400; PA – L50 and L250; VW, MA(s1), MA(s2) and SPA – L50/50, L50/150, L250/50 and L250/150); and “M” or “C” indicates the loading type that the specimens were subjected to (e.g. M – stands for monotonic while C stands for cyclic loading). As an example, model VW-CFRP/Steel-L250/50:C corresponds to the numerical simulation of the adhesively bonded joint between CFRP and steel with a free bonded length of 250 mm followed by an additional anchorage with a CFRP strip with Variable Width with an anchored length of 50 mm, subjected to the cyclic loading protocol.

3.3. Monotonic and cyclic loading protocols

The single-lap shear test was implemented in displacement control mode for both monotonic and cyclic loading protocols. The CFRP pulled end is located at the opposite side of the anchorage, and the pulling loads were simulated through a regular displacement speed of 10 mm/s (with a calculation step of 2.5×10^{-11} s), which was implemented in both monotonic and cyclic loading protocols. This way, at every instant of the simulation, the interfacial slip between the CFRP and the substrate on the CFRP pulled end is the same in all the specimens. In the monotonic test, two failure modes can be obtained. One corresponds to the cohesion failure mode of the adhesive, where the CFRP is pulled out until the total bonded length has fully debonded from the steel substrate or the CFRP has slipped from inside the additional anchorage. The other possible failure mode corresponds to the rupture of the CFRP strip.

Fig. 6 illustrates the implemented cyclic loading protocol, where the maximum slip is increased every three loading cycles up to 2.776 mm. If the adhesively bonded joint is able to sustain the 30 cycles (which did not occur in any of the studied numerical specimens), we planned to apply a monotonic load until failure. The increase of the maximum slip in the CFRP pulled end was adopted to occur at every three consecutive cycles and its increase was assumed to be equal to 0.040 mm in the first four levels, and then the amplitude increase was doubled in each step. Therefore, the loading protocol contemplated the following ten nominated slips of 0.136 mm, 0.176 mm (+0.040 mm), 0.216 mm (+0.040 mm), 0.256 mm (+0.040 mm), 0.296 mm (+0.040 mm), 0.376 mm (+0.080 mm), 0.536 mm (+0.160 mm), 0.856 mm (+0.320 mm), 1.496 mm (+0.640 mm), and 2.776 mm (+1.280 mm). To avoid the application of a compression load during the unloading phase, i.e. pushing the CFRP rather than pulling it, the unloading process was interrupted when the load at the CFRP pulled end (F) was equal to or lower than 250 N.

4. Results and discussion

In this section, the results obtained from the numerical simulations are presented and the findings are discussed. Three main topics are addressed in this section. The first one is dedicated to the influence of the bonded length and only the results obtained from the specimens subjected to the monotonic loading in Series 1 are discussed. Secondly, an analysis of the influence of the additional anchorage on the specimens is carried out. Despite the fact all the series are now considered in this subsection, only the results obtained from the specimens subjected to the monotonic load are reported. Finally, the third subsection is mainly devoted to the analysis of the cyclic bond behaviour obtained from specimens in all series considered in this work. In this third subsection, the load-slip responses of the specimens are presented. The analysis of the damage progress (local or global) obtained from the specimens is another common parameter to all subsections.

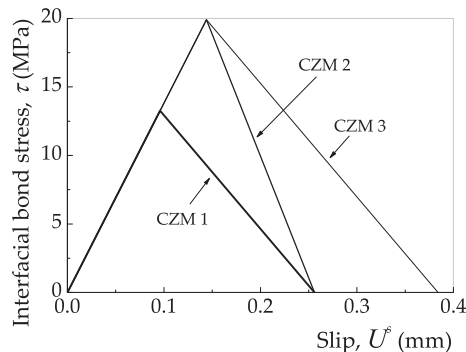


Fig. 5. Cohesive Zone Models (CZM) with a bilinear shape adopted in the numerical specimens.

4.1. Influence of the bonded length

Although this is a well-known parameter with great influence on the bond behaviour of all overlapped bonded joints, regardless of the types of the adherends, it is still ignored in various works available in the literature, e.g. [84–88]. To show how the bonded length influences the bond behaviour of the simulated numerical models, in the subsequent subsections three aspects are analysed thoroughly: (i) global bond behaviour, i.e. the load vs. slip response of the overlapped bonded joints; (ii) the interfacial stresses developed within the overlapped bonded length of the joints; and (iii) local damage D^U evolution under monotonic loading.

4.1.1. Load vs. Slip behaviour of the reference specimens

To analyse the global bond behaviour of the reference bonded joints and make a comparison with local bond behaviour, the load-slip relationships obtained in the simulated numerical specimens from Series 1 are shown in Fig. 7a. The results clearly show that the interfacial strength of the bonded joint increases with the bonded length up to a certain point, and, thereafter, a plateau at maximum load can be observed. The increase of the bonded length has, therefore, an important impact on the ductility of the joint.

The identification of the length beyond which the interfacial strength cannot increase any further, i.e. the identification of the effective bond length (L_{eff}), can be carried out using the global load-slip relationship by identifying the specimens where the previously mentioned plateau with maximum load can be observed. Since in the this case the reference specimen REF-CFRP/Steel-L100:M has almost reached the maximum interfacial strength of the joint, it can be asseverated that the effective bond length should be slightly longer than 100 mm. The calculation of L_{eff} can be carried out from the interfacial strength vs. bonded length relationship (see Fig. 7b). Thus, two different branches can be defined. The first one, defining the trend of the interfacial strength of the joints with short bonded lengths, and the second one representing the interfacial strength of the joints with a sufficiently long bonded length. Unlike the latter branch which is constant, the former is nonlinear and its definition has been the motive of several studies available in the literature, e.g. [89–91]. In the current case, the proposal made by Neubauer and Rostásy [89] was used to estimate the maximum load (or the debonding load of the reference specimens). To that end, the following minimization process was performed:

$$\xi = \min_{L_{eff}} \sum_i^n \sqrt{(\beta_L \cdot F_{deb,i} - F_{max,i})^2} \quad (32)$$

where $F_{max,i}$ and $F_{deb,i}$ are the maximum load of the reference specimens (i.e. Series 1) and the debonding load of the specimens with short bonded lengths (otherwise, its values may correspond to $F_{max,i}$); and the dimensionless parameter β_L is given by [89]:

$$\beta_L = \frac{L_b}{L_{eff}} \cdot \left(2 - \frac{L_b}{L_{eff}} \right) \quad (33)$$

The calculated effective bond length of the specimens is approximately 120 mm (see Fig. 7b) which justifies the option of choosing in this study two different bonded lengths (50 mm and 250 mm) for the anchored specimens (Series 2 to 6). Thus, it is possible to analyse two different situations per anchorage type, i.e. with an initial short and with a sufficient free bonded length.

4.1.2. Interfacial stresses and local damage

The interfacial stress distribution as well as the interfacial local damage (see Eq. (15)) obtained from the EBR specimens (in Series 1) are shown in Fig. 8. The eight different points previously defined for the cyclic loading protocol, i.e. between 0.136 mm and 0.856 mm, were selected to show how the interfacial stresses develop along the overlapped bonded length of the CFRP-to-steel joint. The results showed that the left side of all specimens has softened before the complete debonding of the CFRP from the steel substrate. However, in the case shown in Fig. 8a (REF-CFRP/Steel-L50:M) the full damage of the total bonded length of the specimen is rapidly reached. As the bonded length increases, the elastic and softening interfacial stresses tend to spread over a longer bonded length. The maximum bond stress developed within the CFRP-to-steel interface follows the same direction as the debonding progress, i.e. it moves from the CFRP pulled end towards the CFRP unpulled end (see, e.g., Fig. 8f).

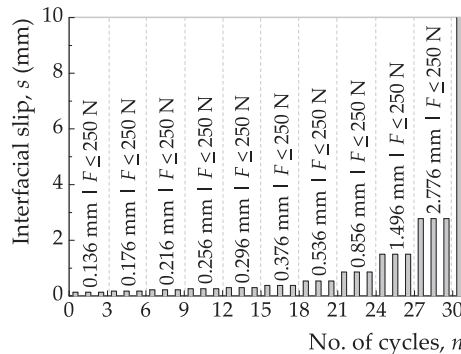


Fig. 6. Variation of the maximum interfacial slip with loading cycles in the numerical simulations.

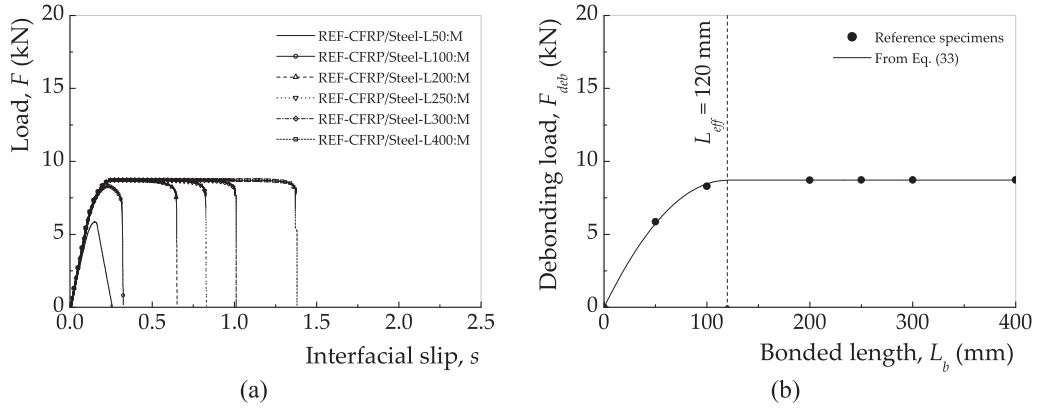


Fig. 7. Global bond performance of the numerical specimens from Series 1: (a) load vs. slip response; and (b) interfacial strength vs. bonded length relationship (identification of L_{eff}).

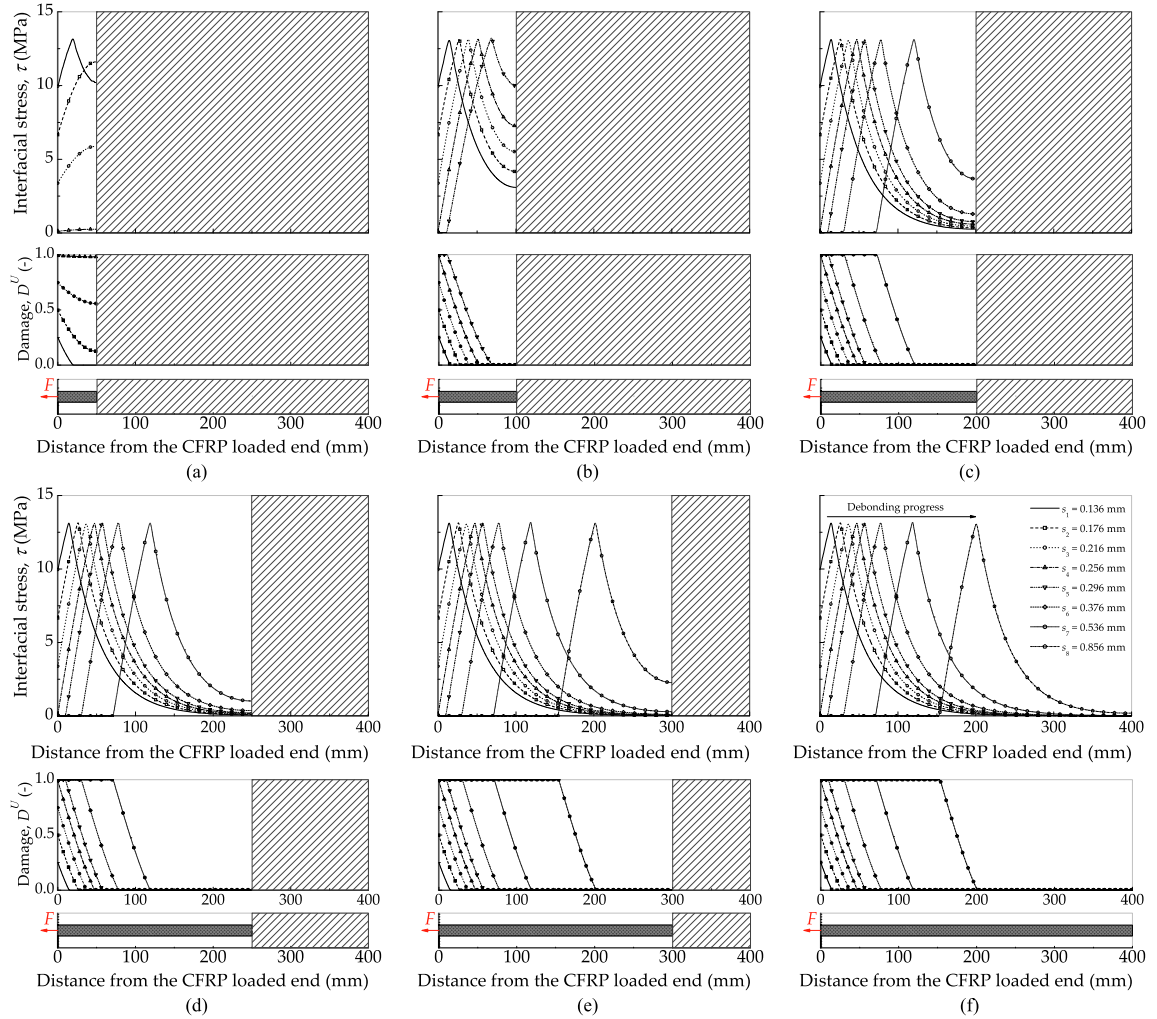


Fig. 8. Interfacial stresses in the numerical specimens of Series 1 under monotonic loading with an overlapped bonded length of: (a) 50 mm; (b) 100 mm; (c) 200 mm; (d) 250 mm; (e) 300 mm; and (f) 400 mm.

The damage of the CFRP-to-steel joint caused by the increase of the load F tends to linearly increase between the Elastic (E) region, where the damage is 0%, and the Debonded (D) region, where the damage reaches its maximum value of 100%. Before the complete debonding, the specimen with the largest bonded length (REF-CFRP/Steel-L400:M) was able to reach the interfacial slip of 0.856 mm with almost one-half of the bonded length fully damaged. Therefore, bearing in mind that the effective bond length of the CFRP-to-steel joint (~ 120 mm) is still shorter than the remaining undamaged length of 200 mm, it can be stated that the load capacity of this specimen would not be affected at this interfacial slip imposed on the bonded joint if, after unloading the specimen, the CFRP-to-steel joint was reloaded.

4.2. Influence of the adopted anchorage

4.2.1. Load vs. Slip behaviour

To analyse the influence that each anchorage has on the global bond performance of the CFRP-to-steel bonded joints, the load-slip curves obtained from the simulations are compared with the homologous specimens in Series 1, i.e. with the reference specimens with the same total bonded length. Therefore, Fig. 9 compares the load-slip responses of the specimens in Series 2 to 6 with the corresponding ones in Series 1. Excluding the comparisons with the specimens in Series 6, Fig. 9 shows that using an additional anchorage with a short length is not an efficient solution since it hardly increases the strength of the bonded joint. On the other hand, when the length of the anchorage increased from 50 mm to 150 mm, the interfacial strength of all specimens in Series 2 to 5 increased as well. In this regard, the anchorage had the most impact on specimen VW-CFRP/Steel-L250/150:M (from Series 2 in Fig. 9a), which showed an increase of 80.5% from its corresponding reference interfacial strength (from 8.72 kN in specimen REF-CFRP/Steel-L400:M to 15.74 kN). It should be mentioned also that the interfacial strength of specimen VW-CFRP/Steel-L50/150:M was almost the same as specimen VW-CFRP/Steel-L250/150:M (i.e. 15.71 kN), which suggests that despite the initial short length of the joint, which is only 50 mm, the strength of the joint is almost entirely comprised in the anchorage (the same length of anchorage adopted in both specimens).

Looking at the specimens in Series 3 and 4 where the mixed adhesive concept was applied, the specimens where the CZM had the highest fracture energy, i.e. in Model 3, provided the highest interfacial strength. Compared with the reference specimens, the interfacial strength of specimens MA(s2)-CFRP/Steel-L50/150:M and MA(s2)-CFRP/Steel-L250/150:M increased 48.4% and 49.2%, respectively (from 8.72 kN to 12.94 kN and 13.01 kN, respectively). Furthermore, the load-slip responses obtained from the specimens in Series 3 and 4 allow us to observe a second plateau at a higher load than that obtained from the reference specimens. As with these reference specimens, and as already explained, this new plateau for the specimens in Series 3 and 4 means that the additional 150 mm with a different CZM is enough to allow the anchorage to reach its maximum strength.

The influence of the mechanical anchorage simulated in the Series 5 specimens is demonstrated in Fig. 9d. The results show that the addition of a (rigid) steel plate to the unpulled end of the CFRP strip increased the strength of the bonded joint up to 12.30 kN and 12.33 kN in specimens SPA-CFRP/Steel-L50/150:M and SPA-CFRP/Steel-L250/150:M, respectively, which means a load increase from their homologous reference specimens of 41.1% and 41.4%, respectively. In these two cases, the load-slip responses show that the CFRP strip slips from inside the anchorage. To prevent this, two main strategies could be followed. One is to use an adhesive that could enhance the bond stress transfer between the steel plate and the CFRP strip. As with the specimens in Series 3 and 4, the adhesives that might be associated with an improved local bond-slip relationship increase the bond performance of the joint. The other way has the

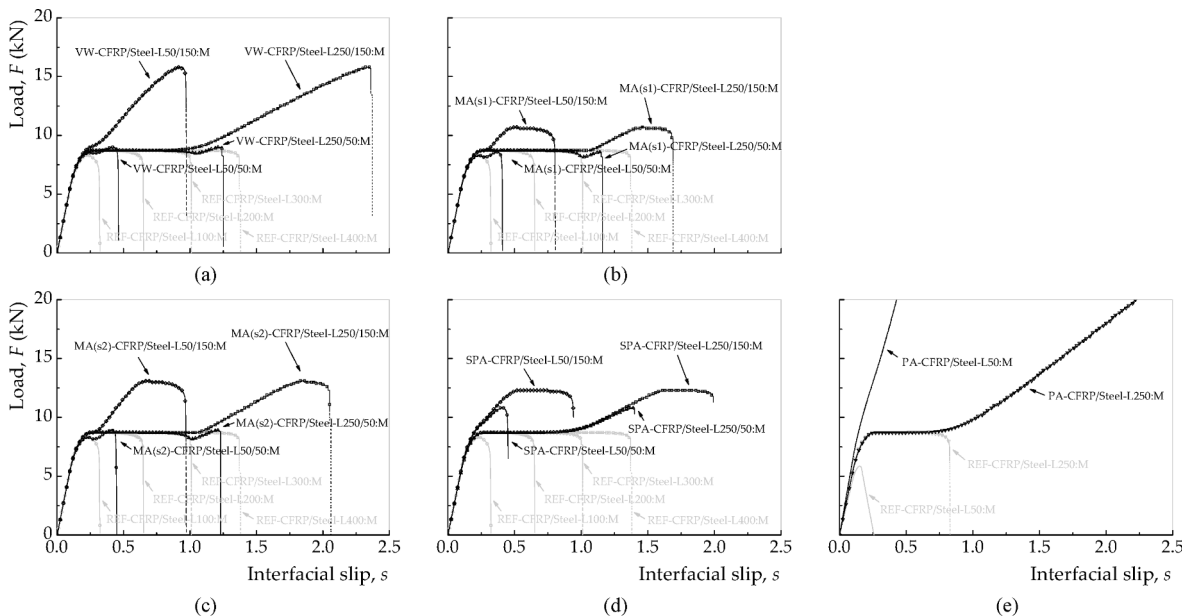


Fig. 9. Monotonic load vs. slip response of the numerical specimens from: (a) Series 2; (b) Series 3; (c) Series 4; (d) Series 5; and (e) Series 6.

same aim, i.e. to enhance the local bond behaviour of the joint, is to apply an external load to the steel plate against the CFRP strip. In fact, this is a common strategy used by several researchers [92–94] in which the amount of pressure that needs to be applied onto the steel plate should be carefully designed to avoid the shear cut of several carbon fibres with an overestimated load pressure and so creating, localized and unnoticed damage of the CFRP composite [83,95].

The load-slip responses of the two specimens in Series 6 shown in Fig. 9e illustrate the “ideal” situation, i.e. they lead to a classic failure mode, which is the cohesive rupture of the CFRP strip whether the bonded length is short or long. The main difference between both lies precisely in the adopted bonded length where the bonded joint which is longer than the effective one (i.e. in specimens PA-CFRP/Steel-L250:M) shows three main distinct branches in its load-slip response. The first two branches of this global behaviour are equal to the load-slip response of the homologous reference specimen because these are attained before the anchorage is loaded

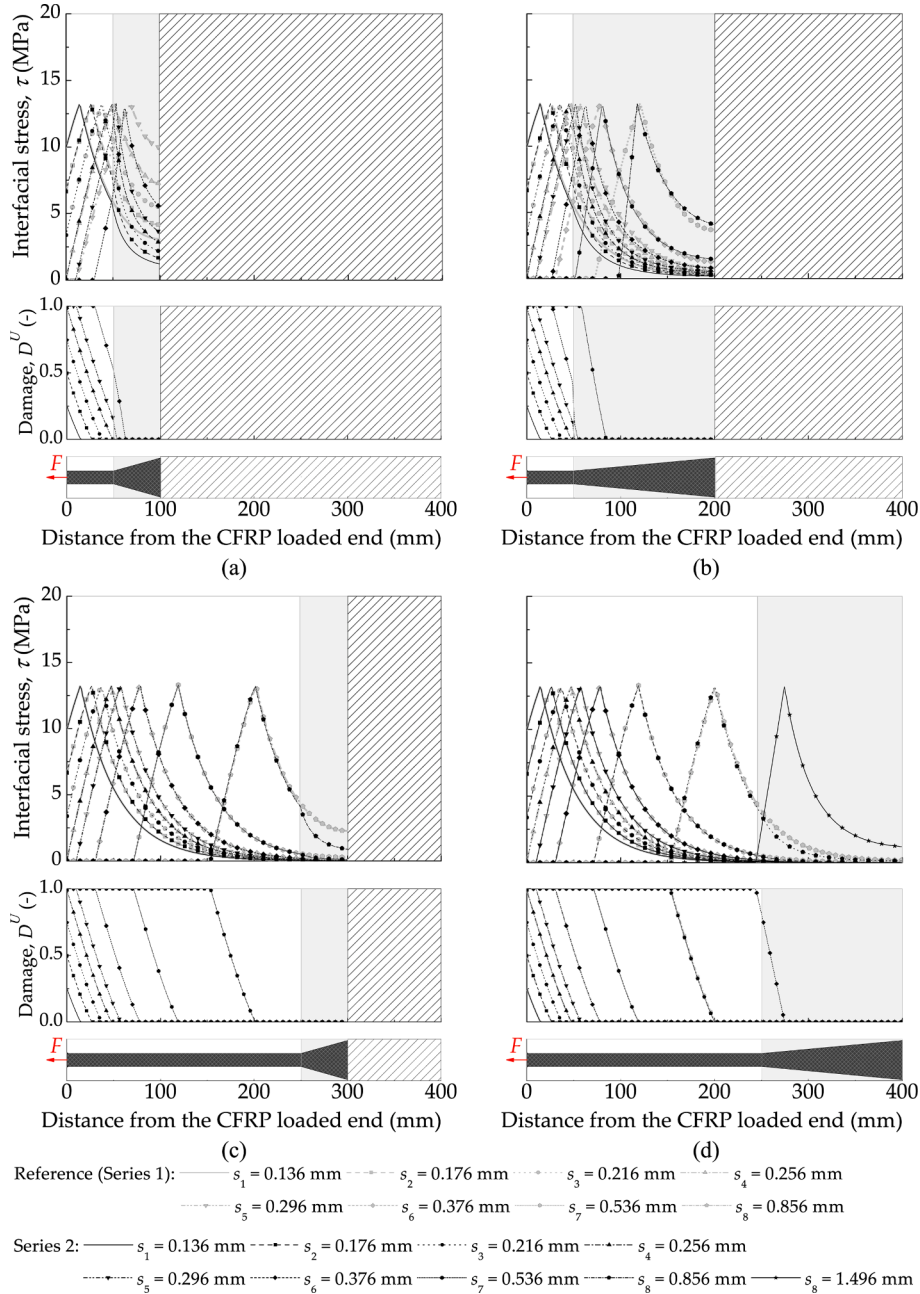


Fig. 10. Interfacial stresses in the numerical specimens of Series 2 under monotonic loading with a total overlapped bonded length of: (a) 100 mm – 50 mm plus 50 mm anchored length; (b) 200 mm – 50 mm plus 150 mm anchored length; (c) 300 mm – 250 mm plus 50 mm anchored length; and (d) 400 mm – 250 mm plus 150 mm anchored length.

(interfacial stress is null). However, as the debonding process moves towards the CFRP unpulled end, the third branch in the load-slip initiates and it tends to a linear response with a stiffness approximately equal to the axial stiffness of the CFRP strip [66,78].

In terms of the interfacial slips, it should be mentioned that the load-slip curves shown in Fig. 9 revealed that the use of an additional anchorage system increases the interfacial slip at the failure of the specimens. However, the most pronounced increases can be seen in specimens VW-CFRP/Steel-L250/150:M and PA-CFRP/Steel-L250:M. In the first of these (see Fig. 9a), the interfacial slip at its failure point reached 2.367 mm which, compared with the result obtained from its homologous reference specimen, is an increase of approximately 71.9%. In the case where the CFRP composite is assumed to be perfectly anchored to the steel (see Fig. 9e), the interfacial slip at failure (CFRP rupture) of specimen PA-CFRP/Steel-L250:M was 2.442 mm, which means that an increase of approximately 194.6% was obtained when compared to the reference specimen with the same total bonded length of 250 mm (i.e. REF-CFRP/Steel-L250:M).

4.2.2. Interfacial stresses and local damage of Series 2 specimens

Fig. 10 shows the interfacial stresses and the corresponding local damage of the specimens with an additional bonded length where the CFRP linearly varied until its unpulled end. The influence of this anchorage is highlighted by comparing the results obtained in the Series 2 specimens (black line curves) to the ones obtained from the reference specimens (in Series 1) (grey line curves) with the same total bonded length. In general, the curves of both series are superimposed for the same slip at the pulled end. The main differences between the specimens in these two series can be observed in the specimens with the shortest additional anchorage, i.e. specimens VW-CFRP/Steel-L50/50:M (see Fig. 10a) and VW-CFRP/Steel-L250/50:M (see Fig. 10c). So, the use of an additional bonded length with a CFRP with variable width, even with a short length of 50 mm, tends to reduce the interfacial bond stresses on the right side of the joint, as shown in Fig. 10a and Fig. 10c. However, since these differences are located in an Elastic (E) region, the calculated damage to the bonded joints in both series has the same distributions along the bonded length in all the four cases illustrated in Fig. 10.

The anchorage did not change the location of the maximum bond stress at the fixed interfacial slip. However, the results obtained from specimen VW-CFRP/Steel-L250/150:M (see Fig. 10c) showed that when the interfacial slip reached 1.496 mm the homologous specimen in Series 1 had already fully debonded and no results could be shown. Therefore, this reveals the gain that can be obtained by varying the width of the CFRP as an additional anchorage since it allows the interface to reach a higher interfacial slip without reaching the complete debonding of the joint.

4.2.3. Interfacial stresses and local damage of Series 3 specimens

The interfacial stresses obtained from the Series 3 specimens as well as their corresponding damages are compared with the reference data in Fig. 11 (colour code: black line curves for Series 3 and grey line curves for the reference). In the specimens in this series, the anchorage length was improved with a stiffer adhesive which could increase the maximum bond stress developed within the CFRP-to-steel bonded joint. Although the Elastic (E) stage of both adhesives had the same interfacial stiffness, the Softening (S) stage of the adhesive adopted for the anchorage decays abruptly until 0.256 mm (i.e. equal to U_{ult}^S), beyond which the detachment between materials is assumed. Under these conditions, the results show that the interfacial stresses within the CFRP-to-steel bonded joints have significantly changed from their homologous specimens. Nevertheless, such differences are clearer in the specimens with an anchorage length of 150 mm (see Fig. 11b and 11d). Particularly in specimen MA(s1)-CFRP/Steel-L50/150:M (see Fig. 11b) and despite having the same interfacial slip, the interfacial stresses corresponding to the E and S regions have completely different configurations from its homologous specimen REF-CFRP/Steel-L200:M.

Regarding the damage done to the CFRP-to-steel bonded joint, there are some important differences from the description in the previous section. For instance, looking at the damage evolution of specimens MA(s1)-CFRP/Steel-L50/50:M and MA(s1)-CFRP/Steel-L50/150:M, when the first 50 mm are in the S stage and the anchorage enters in it also, the damage stops increasing linearly and a discontinuity can be observed between 0% (undamaged) and 100% (fully damaged). The second aspect can be seen from the results obtained from specimen MA(s1)-CFRP/Steel-L250/150:M (see Fig. 11d). The damage curves between 0% and 100% are parallel to each other, as the CFRP debonds from the steel substrate. However, when the 250 mm are fully debonded and only the anchorage can sustain the loads transmitted to the CFRP composite, the damage transition between 0% and 100% happens on a steeper slope. In other words, by increasing the stiffness k_d^S in the bond-slip model, full damage caused to the CFRP-to-steel interface is obtained rapidly. Therefore, this reveals how important the S stage of the bond-slip model can be to preserve the integrity of the joint.

4.2.4. Interfacial stresses and local damage of Series 4 specimens

Fig. 12 shows the interfacial stresses and corresponding damage evolutions obtained in the Series 4 specimens. When compared with the reference specimens, the main differences can be observed when a short bonded length of 50 mm is used. Since the maximum bond stress within the anchorage length increased up to 19.91 MPa the interfacial stresses within these regions were able to develop that peak stress value (see Fig. 12a or b). However, the corresponding damage evolutions of these two specimens (MA(s2)-CFRP/Steel-L50/50:M and MA(s2)-CFRP/Steel-L50/150:M) were not affected and the transition from an undamaged region to a fully damaged region is similar to the reference specimens.

In Fig. 12c, the interfacial stresses developed in the specimen MA(s2)-CFRP/Steel-L250/50:M have no visible differences from that determined in its corresponding reference specimen. On the other hand, the results shown in Fig. 12b or d reveal that adopting a long anchorage makes it possible to take full advantage of using a stiffer and more ductile adhesive, i.e. it is possible to explore the higher maximum bond stress promoted by the adhesive as well as its softening stage which, from the perspective of equilibrium, will increase the bond capacity of the CFRP-to-steel joint. Since the interfacial stiffness in the softening stage of the anchorage length is the same as that used in the free bonded length, the damage evolution has undergone no change.

4.2.5. Interfacial stresses and local damage of Series 5 specimens

In this case, an additional interface was created with the use of a steel plate placed on top of the CFRP-to-steel joint. For this reason, Fig. 13 shows two different bond stress distributions. One, at the top, corresponds to the interface between the CFRP composite and the mechanical anchorage, and the other, on the bottom, corresponds to the interface between the CFRP composite and the steel substrate. Since the mechanical anchorage and the substrate were made of steel, the same bond-slip model was used to locally simulate both interfaces. Therefore, almost no differences from the reference specimens can be identified. However, as in the Series 2 specimens, where the main differences were found in the specimens with the shortest anchorage length, specimens SPA-CFRP/Steel-L50/50:M (see Fig. 13a) and SPA-CFRP/Steel-L50/150:M (see Fig. 13c) also developed lower bond stresses than that developed in their homologous specimens within the mechanical anchorage length.

It should be highlighted also that the additional interface in these specimens, i.e. between the mechanical anchorage and the CFRP

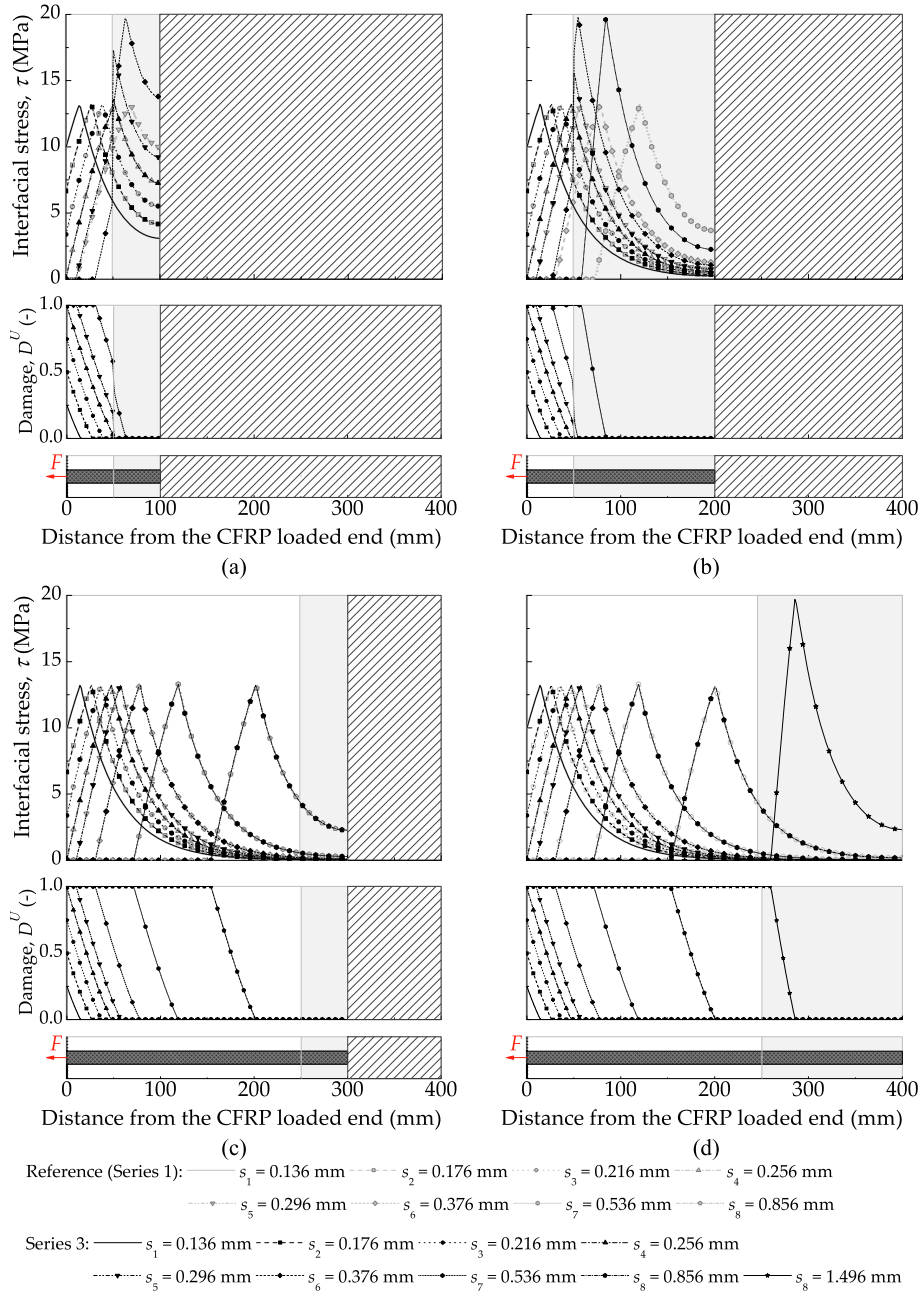


Fig. 11. Interfacial stresses in the numerical specimens of Series 3 under monotonic loading with a total overlapped bonded length of: (a) 100 mm – 50 mm plus 50 mm anchored length; (b) 200 mm – 50 mm plus 150 mm anchored length; (c) 300 mm – 250 mm plus 50 mm anchored length; and (d) 400 mm – 250 mm plus 150 mm anchored length.

composite, is the key aspect for the increase of the load transmitted to the CFRP composite when compared with the reference specimens. So, from the point of view of equilibrium, the interfacial bond stress shown at the top of Fig. 13a to 13d contributes to an additional load capacity of the joints shown in Fig. 9d. However, since the interfacial stresses developed within the mechanical anchorage length are not zero anywhere, it is expected that these specimens will have an interfacial failure either between the mechanical anchorage and the CFRP composite or between the CFRP composite and the steel substrate.

Looking at the damage evolution in both interfaces it can be asseverated that the evolution is the same as that obtained in the reference specimens. This can be explained by the adopted bond-slip model (Model 1 in Fig. 5) that was used to locally simulate all interfaces, whether in the reference specimens (in Series 1) or the anchored specimens in this Series 5.

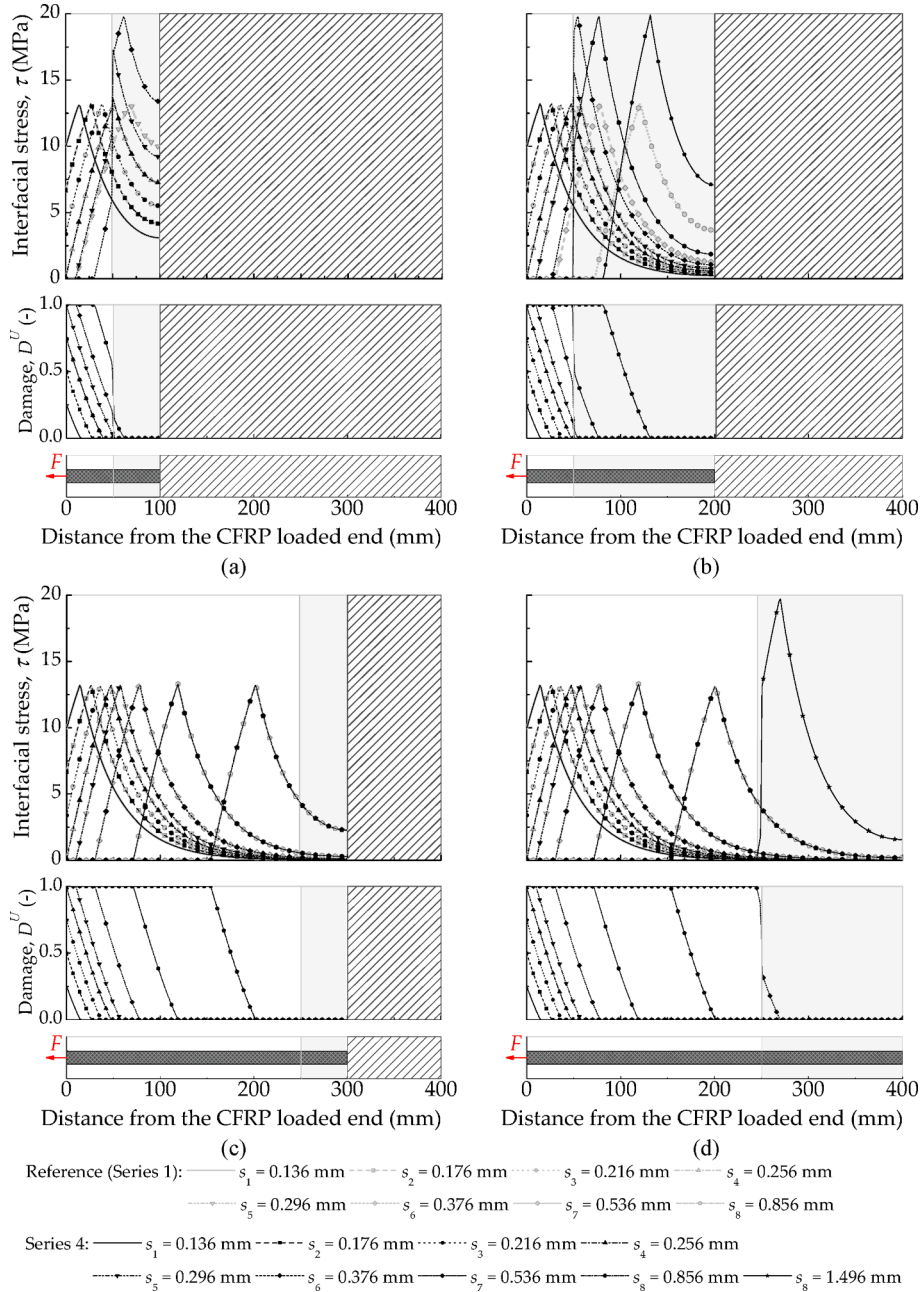


Fig. 12. Interfacial stresses in the numerical specimens of Series 4 under monotonic loading with a total overlapped bonded length of: (a) 100 mm – 50 mm plus 50 mm anchored length; (b) 200 mm – 50 mm plus 150 mm anchored length; (c) 300 mm – 250 mm plus 50 mm anchored length; and (d) 400 mm – 250 mm plus 150 mm anchored length.

4.2.6. Interfacial stresses and local damage of Series 6 specimens

Here the interfacial bond stresses have a different configuration from the other specimens analysed so far. The main difference can be seen at the CFRP unpulled end, i.e. at 50 mm (see Fig. 14a) and 250 mm (see Fig. 14b) from the CFRP loaded end. At these points, once the interfacial slips are constrained, the interfacial bond stresses should be always equal to zero during the full debonding process.

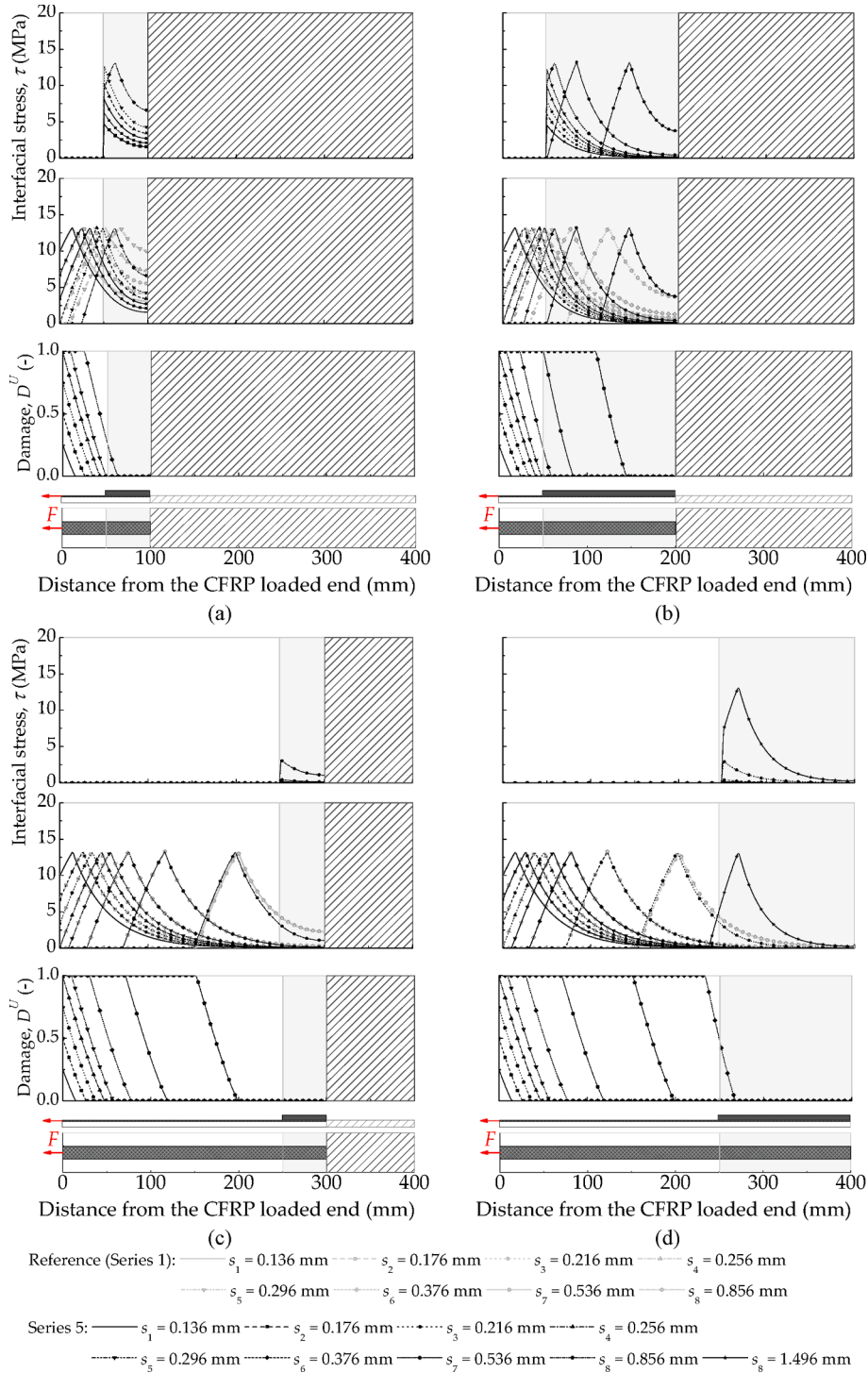


Fig. 13. Interfacial stresses in the numerical specimens of Series 5 under monotonic loading with a total overlapped bonded length of: (a) 100 mm – 50 mm plus 50 mm anchored length; (b) 200 mm – 50 mm plus 150 mm anchored length; (c) 300 mm – 250 mm plus 50 mm anchored length; and (d) 400 mm – 250 mm plus 150 mm anchored length.

Consequently, the configurations of the bond stresses of specimens PA-CFRP/Steel-L50:M (in Fig. 14a) and PA-CFRP/Steel-L250:M (in Fig. 14b) in the vicinity of the CFRP unpulled end are completely different from their homologous specimens REF-CFRP/Steel-L50:M and REF-CFRP/Steel-L250:M, respectively.

As the length of the bond stress diagram decreases, the evolution of the interfacial damage increases. This can be seen from Fig. 14b where the slope of the damage corresponding to $U^f = 1.496$ mm (i.e. the curve on the far right-hand side) has increased when compared with the previous nominated interfacial slips. However, it should be kept in mind that although the damage to the CFRP-to-steel joint occurs rapidly at the latter stage of its debonding process, this is achieved with a high load transmitted to the CFRP composite as can be seen from Fig. 9e.

4.3. Influence of the cyclic loading

In the following section, the influence of the cyclic loading on the different specimens numerically simulated in this study is analysed. To this end, the load-slip response of the specimens is analysed. Moreover, and based on the load-slip response, the evolution of the stiffness during the cyclic loading, and the evolution of the global damage of the specimens are presented to demonstrate how the different anchorages may influence the cyclic bond performance of the joints. The dissipated energy of the CFRP-to-steel bonded joints per cycle as well as the corresponding cumulative energy are two other parameters that are reported in this subsection.

4.3.1. Load vs. Slip behaviour

The influence of cyclic loading on the load-slip response of the different specimens in Series 1 to 5 is shown in Fig. 15. In each graph the ratio of the load reached in each cycle to the monotonic load at the same interfacial slip is also shown. Like the specimens under monotonic loading, the specimens where the adopted anchorage was 150 mm had the best load-slip performance since the load capacity in those specimens was enhanced and, as already mentioned, their load capacity remained unchanged. For this reason, and for the sake of simplicity, in the following analyses and from here after, only the specimens with the longest anchorage are discussed. In this group of specimens with an additional anchorage, like its corresponding reference specimens, specimen VW-CFRP/Steel-L250/150:C (Fig. 15b, right) stands out from the others by reaching the highest load capacity and the highest interfacial slip at failure. However, since the maximum load obtained in this specimen did not reach the corresponding rupture load of the CFRP, an interfacial failure mode is associated with this specimen. All the specimens in Series 1 to 5 were all characterized by an interfacial failure of the CFRP-to-steel, whereas the specimens shown in Fig. 16, since the corresponding rupture of the CFRP was reached, a cohesive rupture of the CFRP was obtained in both cases, i.e. with a bonded length of 50 mm (see Fig. 16a) and 250 mm (see Fig. 16b).

It is worth mentioning that although the first cycles of cyclic loading have degraded the loads transmitted to the CFRP strips, the

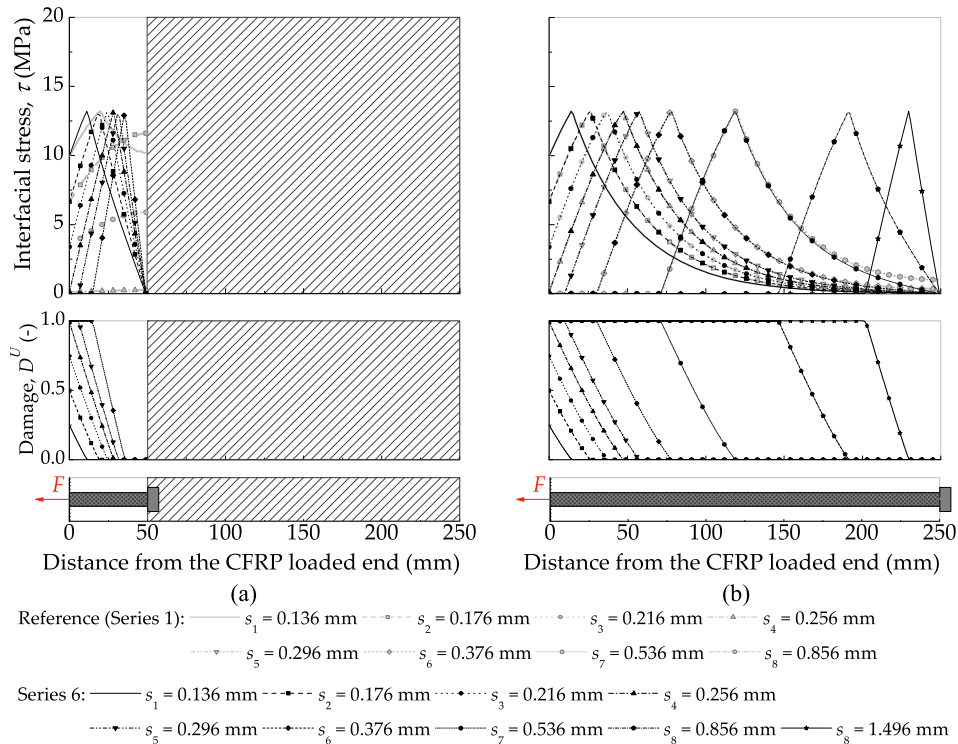


Fig. 14. Interfacial stresses in the numerical specimens of Series 6 under monotonic loading with a total overlapped bonded length of: (a) 50 mm; and (b) 250 mm.

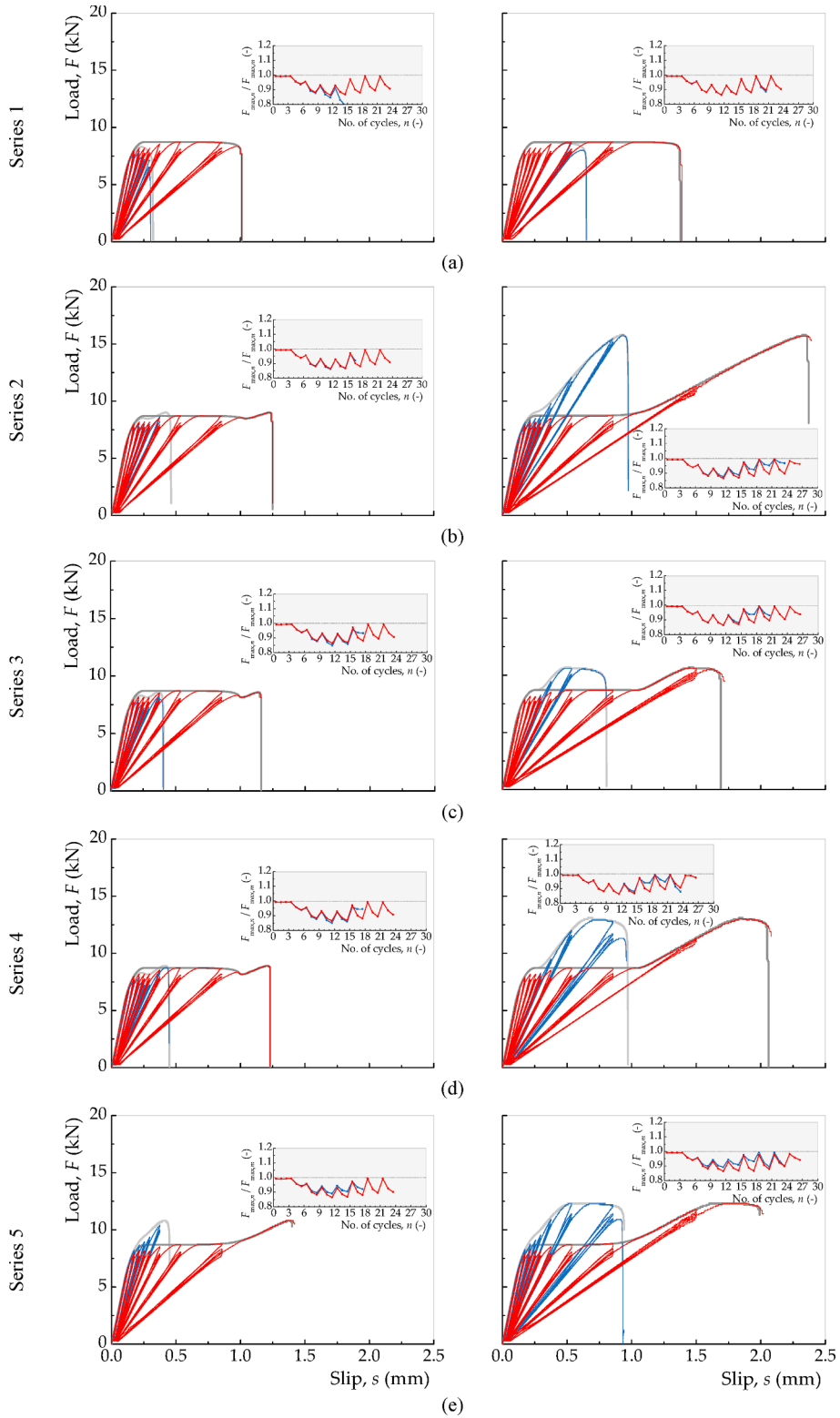


Fig. 15. Cyclic load vs. slip response comparison of the numerical specimens from: (a) Series 1; (b) Series 2; (c) Series 3; (d) Series 4; and (e) Series 5.

joints with the longest bonded lengths reached the same monotonic load transmitted to the CFRP strip for a high number of cycles applied. In general, between the 3rd and the 5th interfacial slip level (i.e. 0.216 mm, 0.256 mm, and 0.296 mm, corresponding to cycles 7 to 15), the load reached in the specimens was lower than that reached in the same specimens subjected to the monotonic load. This shows that despite the fact the CFRP-to-steel bonded joints are being damaged with the number of cycles, the final load capacity may remain unchanged. This can be explained regarding the equilibrium of the CFRP composite in the joint. In other words, as the interfacial bond stresses move towards the CFRP unpulled end, the total bonded length is sufficient to completely install the interfacial bond stress diagram even at the highest number of loading cycles applied.

In addition, for the same slip magnitude of the adopted loading protocol, the second and the third loading cycles always registered a reduction of the load reached in the first loading cycle. This also reveals that for the same slip magnitude, the bonded joint is being damaged with the adopted cyclic loading protocol and can dissipate some energy per cycle, as will be shown later.

4.3.2. Effect of the mechanical anchorage

To complement the analysis of the effect of the mechanical anchorage on the response to the adopted cyclic loading, the evolution of the stiffness and debonded length in the numerical tests were evaluated. Hence, to analyse the evolution of stiffness during the debonding process the stiffness ratio in cycle n , $K_{d,n}^*$, was defined as follows:

$$K_{d,n}^* = \frac{K_{d,n}}{K_e} \quad (34)$$

where $K_{d,n}$ and K_e are the damaged and elastic stiffnesses in cycle n , respectively, calculated according to:

$$K_{d,n} = \frac{\Delta F_n}{\Delta U_n^s} \quad (35)$$

and [96]

$$K_e = b_f \cdot \sqrt{\frac{\tau_{max} \cdot E_f \cdot t_f}{U_m}} \quad (36)$$

where ΔF_n is the variation of the loads transmitted to the CFRP in the unloading/reloading path; ΔU_n^s the variation of the slips in the same unloading/reloading path of ΔF_n ; b_f and t_f are the width and the thickness of the CFRP strip.

From Fig. 17, it can be seen that the stiffness ratio tends to decrease with the number of cycles in all specimens. However, when compared with the results obtained from the reference specimens, the stiffness ratio of the specimens with a free bonded length of 50 mm has a smoother decay for the early loading cycles. On the other hand, the evolution of the stiffness ratio with the number of cycles determined in the specimens with a free bonded length of 250 mm is quite similar to those obtained from their homologous specimens. Still, since the anchorage in each specimen increased the strength of the bonded joint, the calculated stiffness ratio in these anchored specimens continued to decrease whilst the stiffness ratio in the reference specimens suddenly dropped to zero due to the failure of the joint. The difference between the reference (Series 1) and the specimens with an additional anchorage (from Series 2 to 6) is highlighted in Fig. 17. The most significant difference between the anchored specimens and their homologous specimen can be seen in the upper part of Fig. 17e. In this case, the influence of a perfect anchorage on the bonded joint is reflected in the evolution of the stiffness ratio in the first loading cycle and due to the increase of its load capacity, the decay of the stiffness ratio is much smoother than in its corresponding reference specimen.

Another parameter that can be used as an indicator of the influence of the anchorage on the debonding process of the hybrid bonded

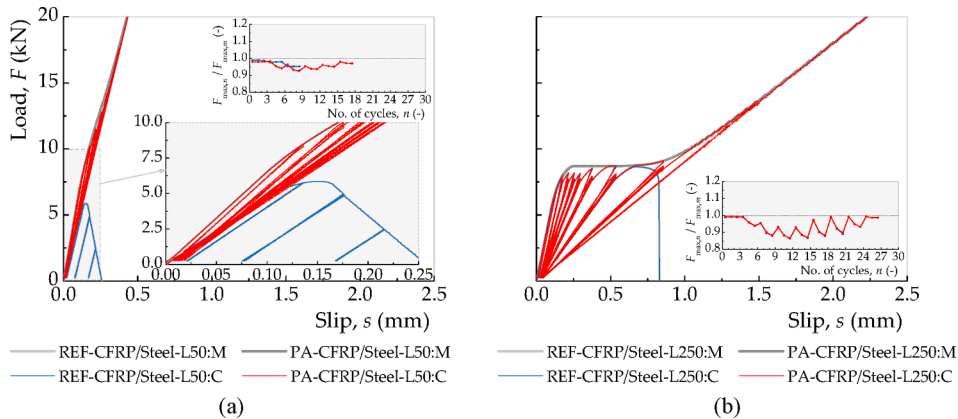


Fig. 16. Cyclic load vs. slip response comparison between the reference specimens (Series 1) with the perfectly anchored specimens (Series 6) with a bonded length of: (a) 50 mm; and (b) 250 mm.

joint subjected to cyclic loading is the evolution of the global damage associated with the specimen as defined in Eq. (16). From a practical point of view, this global damage represents the ratio between the softened and debonded lengths over the total bonded length. Considering a sufficiently long total bonded length, its increase means that the debonded length of the specimen increases as well. Therefore, this parameter indicates how much a specimen can preserve a high percentage of bonded length before failing, without losing its ability to transfer bond stresses for a high number of cycles. To analyse the evolution of the global damage of the CFRP-to-steel bonded joints during their debonding processes, the bond stress distributions developed in all the specimens were monitored in each loading cycle.

In addition to the results obtained for the evolution of the stiffness ratio, Fig. 17 also shows the evolution of the global damage obtained for each specimen with an additional anchorage of 150 mm. Once again, the results obtained from the specimens in Series 2 to 6 are compared to the results obtained from their homologous specimens in Series 1 (reference). Unlike the stiffness ratio, the evolution of the global damage increases with the number of cycles, as would be expected since the debonded length tends to increase during the debonding process of the CFRP-to-steel joint. The evolution of this parameter also shows that the specimens with an additional anchorage and with a free bonded length of 50 mm diverge from the evolution of their homologous reference specimens at a certain point (or number of cycles). Nevertheless, this point does not always coincide with the point previously identified for the evolution of the stiffness ratio (see the upper graphs in Fig. 17a, d, or e). Another interesting aspect that can be seen from Fig. 17, is that the evolution of the global damage is smoother in the specimens with an additional anchorage than that calculated from the corresponding reference specimens. In this particular case, the smallest difference can be found in the specimens MA(s1)-CFRP/Steel-L50/150:C (i.e. from Series 3), which means that the adopted anchorage in this specimen had almost no influence on the debonding process of the joint.

To quantify the deviation of the evolution of the normalized stiffness and the global damage, the Mean Relative Percent Deviation Modulus (MRPDM) was determined according to:

$$d_m = \frac{1}{N} \cdot \sum_{n=0}^N \frac{|v_n - v_n^{REF}|}{v_n^{REF}} \quad (37)$$

where v_n and v_n^{REF} are the values obtained from the specimens with an additional anchorage for the CFRP composite and the reference

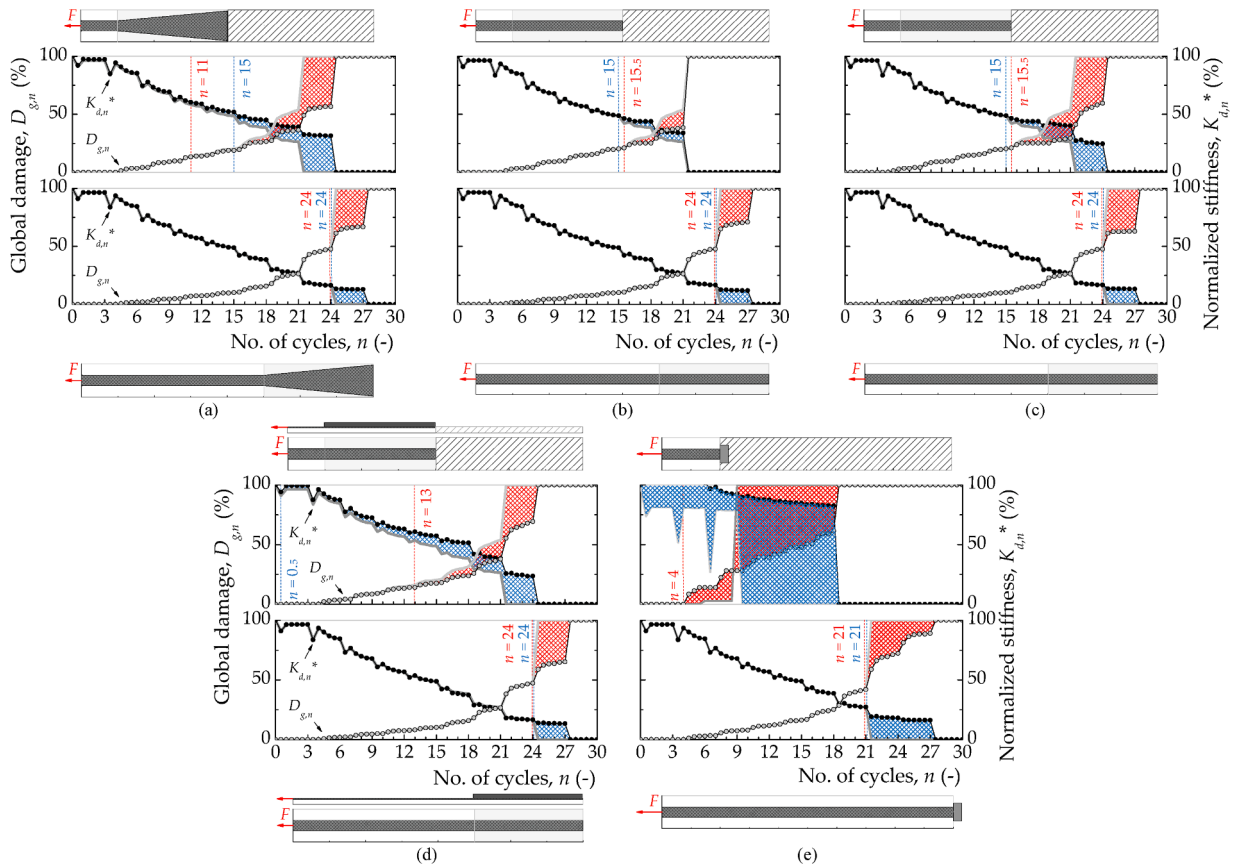


Fig. 17. Comparisons between the normalized stiffness in Eq. (35) and the global damage in Eq. (16) with the specimens in Series 1 throughout the number of cycles carried out during the simulations of the specimens in: (a) Series 2; (b) Series 3; (c) Series 4; (d) Series 5; and (e) Series 6.

specimens, respectively; and N is the maximum number of cycles reached in one of the two specimens being compared. The results are shown in Fig. 18 and in the case of the previously mentioned specimens, the mean deviations of the normalized stiffness and global damage of the reference specimen (REF-CFRP/Steel-L200:C) from specimen MA(s1)-CFRP/Steel-L50/150:C are 3.5% and 3.6%, respectively. In contrast, the largest differences can be seen from specimens SPA-CFRP/Steel-L50/150:C (i.e. from Series 5) and PA-CFRP/Steel-L50:C (i.e. from Series 6). So, for specimen SPA-CFRP/Steel-L50/150:C, the calculated mean deviations for the normalized stiffness ratio and global damage are 11.1% and 9.9%, respectively, whilst specimen PA-CFRP/Steel-L50:C gave mean deviations of 13.9% and 77.9%, respectively. These results clearly show how the perfect anchorage can positively affect the debonding process of a CFRP-to-steel bonded joint with a small bonded length.

4.3.3. Dissipated energy

A bonded joint with a superior ability to dissipate energy is a joint that can better resist cyclic loading without collapse. Although this may be related to the overlapped bonded length, the dissipated energy is a parameter that may provide some important information about the integrity of the bonded joint after facing a cyclic loading such as, for instance, a fatigue load. To analyse the ability of each specimen series to dissipate energy during the cyclic loading, the dissipated energy obtained for each cycle and specimen was calculated according to the following equation:

$$E_{d,n} = \int_{U_{n-1}^s}^{U_n^s} F dU^s \quad (38)$$

where U_{n-1}^s and U_n^s are the interfacial slips at consecutive cycles $n-1$ and n , respectively, measured at the CFRP pulled end.

At the same time, the cumulative dissipated energy was calculated by adding the dissipated energies determined per cycle to Eq. (38). The results are shown in Fig. 19, in which the anchored specimens are compared with their homologous reference specimens in Series 1. The results clearly show that all the specimens tend to dissipate a higher amount of energy during the first cycle with the same interfacial slip than in any of the other two consecutive cycles.

Considering only the specimens with a total bonded length of 400 mm, the results show that these specimens have a greater ability to dissipate energy with more incidence in the later loading cycles. The differences from the reference specimens are almost nonexistent and the dissipated energy, as well as their cumulative values, have not undergone meaningful changes during the number of cycles. Nevertheless, due to the influence of the anchorage, specimen MA(s1)-CFRP/Steel-L250/150:C (i.e. in Series 3 at the lower part of Fig. 19b), was the specimen where the highest cumulative dissipated energy was determined, reaching an approximate value of 6.0 J. In contrast, the specimen with the CFRP perfectly anchored to the steel substrate, i.e. specimen MA(s1)-CFRP/Steel-L250/150:C, had the lowest cumulative energy (see the lower part of Fig. 19e), reaching a value of approximately 4.0 J. This shows that using an additional anchorage with the ability to perfectly restrain the interfacial slips may affect its ability to dissipate energy during cyclic loading, and consequently, reduce its efficiency to face a tougher cyclic loading scenario. In this case, although an interfacial failure can be avoided, the rupture of the CFRP composite is rapidly reached.

Looking now at the specimens with a total bonded length of less than 200 mm, some differences between the anchored specimens and the reference ones can be seen in Fig. 19. For instance, with the exception of specimen PA-CFRP/Steel-L50/50:C, the dissipated energies obtained from all the others anchored specimens have a lower value than that obtained from the reference specimens. So, unless the load capacities of these anchored joints with a free bonded length of 50 mm are increased, the results suggest that the use of an anchorage seems to affect the joint by reducing its ability to dissipate energy during a cyclic loading scenario. Specimen MA(s2)-CFRP/Steel-L50/150:C (see the upper graph in Fig. 19c) is a good example of this. For instance, the increase of the load capacity of this specimen (see the right-hand side of Fig. 15d), allowed us to calculate a dissipated energy value of approximately 2.0 J in the 22nd cycle, which was the maximum value determined in all the specimens in one single cycle. With a lower value of approximately 1.8 J, specimen SPA-CFRP/Steel-L50/150:C (see the upper graph in Fig. 19d), was another specimen where a significant amount of dissipated energy was calculated mainly due to the influence of the steel plate anchorage.

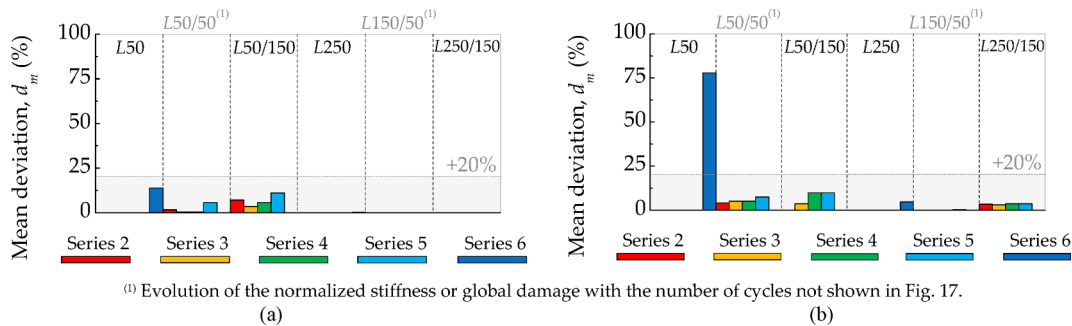


Fig. 18. Mean deviations obtained between the specimens with an additional anchorage and their homologous reference specimens regarding the: (a) normalized stiffness ratio; and (b) global damage.

4.3.4. Overall analysis of the results

The discussion in the previous sections showed that the evaluation of the performance of adhesively bonded joints is complex due, e.g., to the nature of materials used, the loading protocol type, or the bond properties between the adherends used to evaluate the adhesively bonded structure. As the CFRP-to-steel bonded joints with an additional short anchorage length were shown to be inefficient, the results of the specimens with the longest anchorage length (150 mm) were discussed in detail for Series 2 to 5 as well as for both specimens in Series 6. To facilitate the identification of which parameter had the greatest impact on each type of specimen, this section gives an overview of all the six series of specimens studied in this work. Therefore, briefly, Fig. 20 shows the bond behaviour obtained for the specimens with the longest anchorage length used to anchor the CFRP to the steel substrate based on five different indicators and compares those results with their homologous reference data, i.e. with the reference specimen with the same total bonded length. The five indicators used here are: (i) the maximum load of the bonded joint (F_{max}); (ii) the interfacial slip at the failure point of the bonded joint (U_{fail}^s); (iii) the stiffness ratio ($100 - K_{d,n}^*$) in the same cycle of the interfacial failure of the reference specimens and the cycle before the interfacial failure of the specimens with an additional anchorage of the CFRP composite; (iv) the global damage ($D_{g,n}$) in the same cycle of the interfacial failure of the reference specimens and the cycle before the interfacial failure of the specimens with an additional anchorage; and (v) the cumulative dissipated energy ($\Sigma E_{d,n}$). For all indicators, the individual performance of the joint relative to each indicator increases with the value determined, i.e. performance is best when the point is further away from the centre of the plot.

Generally, the longest specimens, i.e. with a total length of 400 mm (in Series 2 to 5), reached the highest loading cycle at failure with an associated cohesive failure mode in the adhesive. Specimen PA-CFRP/Steel-L250:C (on the right-hand side of Fig. 20e), reached the highest values on almost all of the five parameters herein reported, except for the cumulative dissipated energy. However, compared with its reference specimen, this parameter increased approximately twice its value. Relatively speaking, the specimen with the shortest total length in Series 6 has registered the highest load and global damage evolution. At the same time, this specimen was able to stand twice the number of loading cycles ($n = 18$) than those obtained in its homologous EBR specimen ($n = 9$). However, like the specimen with a free bonded length of 250 mm, and despite reaching the rupture of the CFRP, it had a deficient amount of dissipated energy during its debonding process, which is the main disadvantage found in both Series 6 specimens.

The main advantage of adopting a specimen like those in Series 2 with a long total bonded length is the high interfacial slip at failure that it allows the adhesively bonded structure to achieve (see the right-hand side of Fig. 20a). However, the length of its free bonded length is quite important to obtain such a result. With a short free bonded length, the interfacial slip at failure in the Series 2

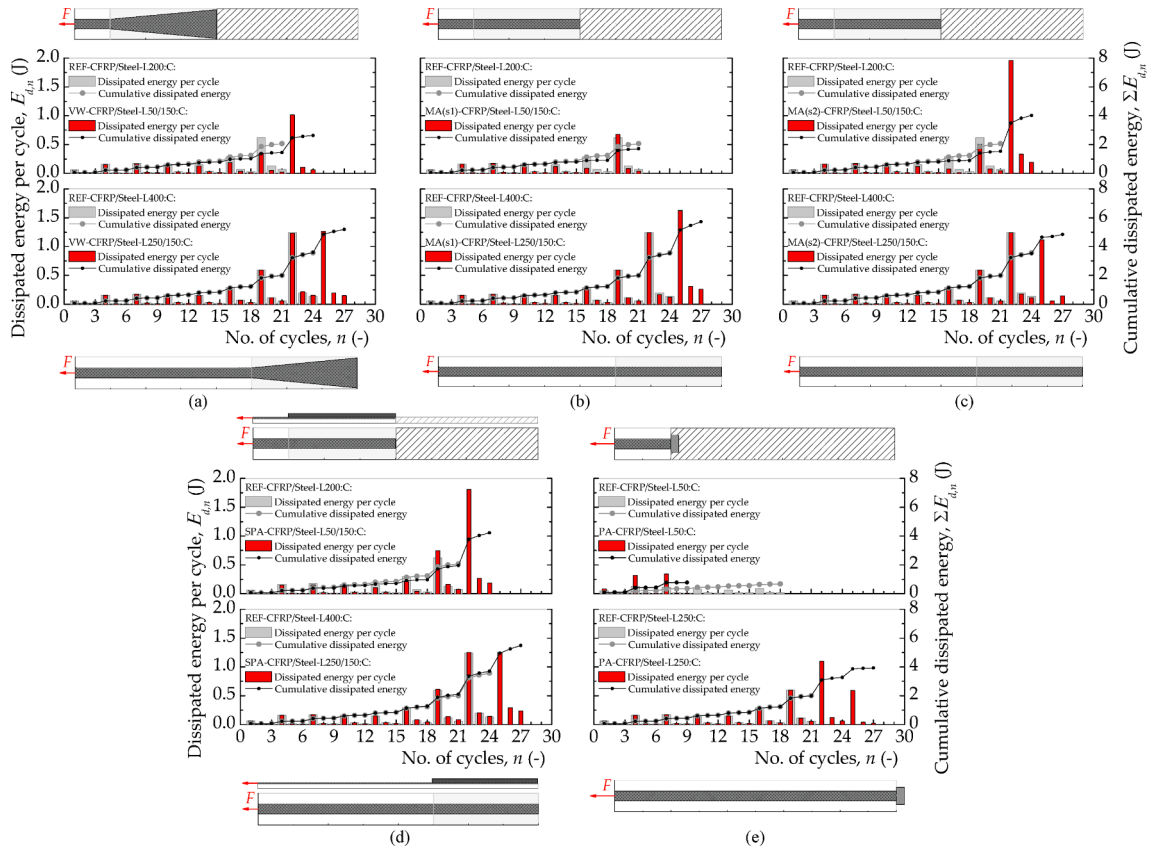
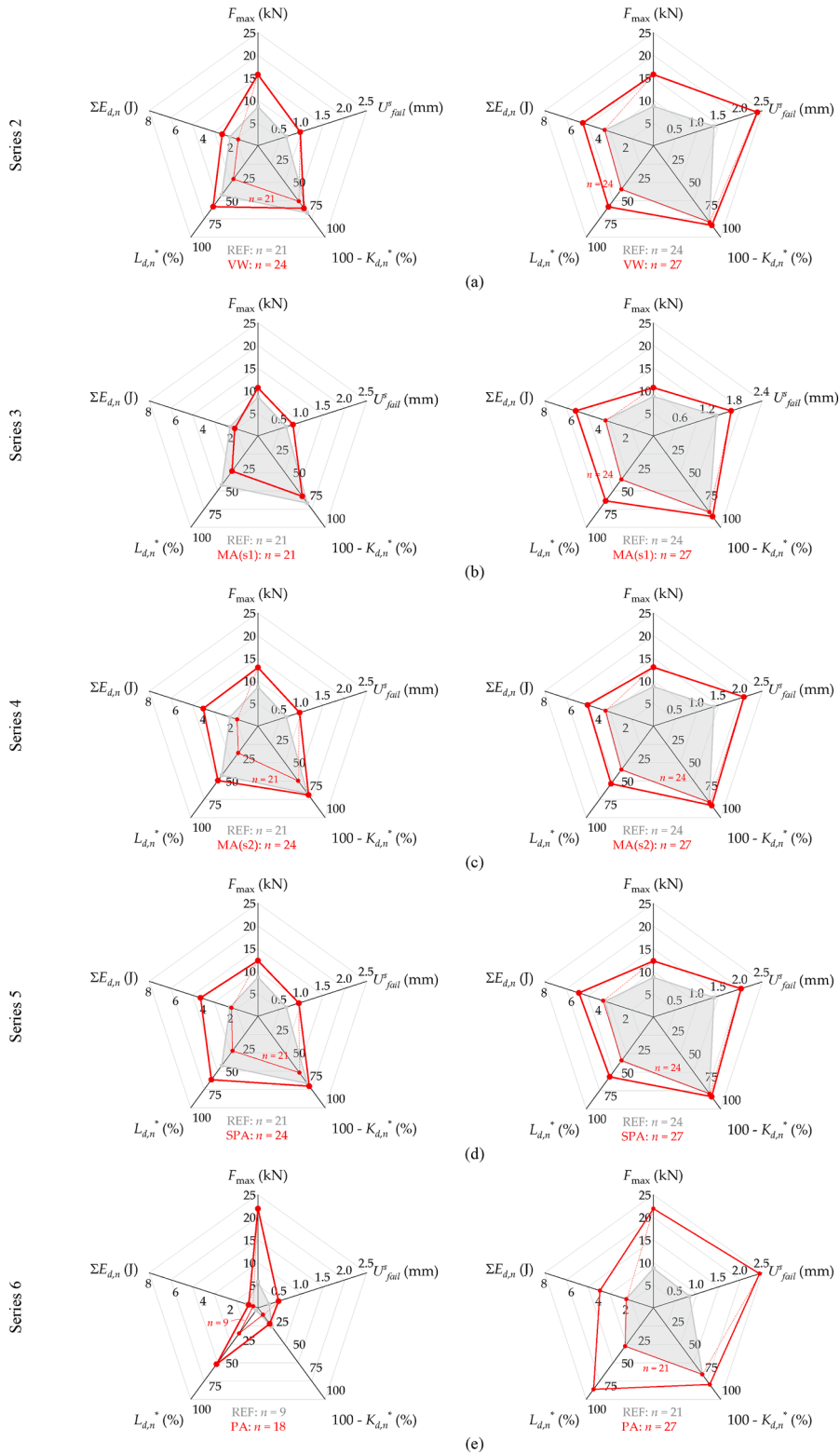


Fig. 19. Comparisons between the dissipated and cumulative energies obtained from the specimens in Series 1 throughout the number of cycles carried out during the simulations of the specimens in: (a) Series 2; (b) Series 3; (c) Series 4; (d) Series 5; and (e) Series 6.



(caption on next page)

Fig. 20. Parametric comparison between the reference specimens in Series 1 and their corresponding specimens with an additional anchorage: (a) Series 2 with $L_t = 200$ mm (on the left-hand) and with $L_t = 400$ mm (on the right-hand); (b) Series 3 with $L_t = 200$ mm (on the left-hand) and with $L_t = 400$ mm (on the right-hand); (c) Series 4 with $L_t = 200$ mm (on the left-hand) and with $L_t = 400$ mm (on the right-hand); (d) Series 5 with $L_t = 200$ mm (on the left-hand) and with $L_t = 400$ mm (on the right-hand); and (e) Series 6 with $L_t = 50$ mm (on the left-hand) and with $L_t = 250$ mm (on the right-hand).

specimen decreases significantly. The other parameter meaningfully affected by the decrease of the free bonded length is the cumulative dissipated energy, which proves that this parameter is highly controlled by the free bonded length adopted for the specimen rather than the length adopted for the additional anchorage.

Like the specimens in Series 2, the different configurations used in the Series 3 specimens led to a poorer bond performance of the specimen with the shortest free bonded length. In this case, the loading cycle reached in this specimen was the same as that reached in its homologous reference EBR specimen (see the left-hand side of Fig. 20b). Looking at the specimen with the shortest free bonded length in Series 4, when the bond-slip relationship was altered from CZM 2 to CZM 3, all five parameters improved. This reveals the importance of the selection of the adhesive used to anchor the CFRP composite to the steel substrate when the CFRP-to-steel bonded joint has a short free bonded length. Hence, the maximum bond stress that the adhesive can provide the CFRP-to-steel joint is not the only condition that should be ensured but how the adhesive softens should also be considered, so as to lead preferably to a high ultimate interfacial slip.

The results obtained from the specimens in Series 5 show that this bonding technique may be a good alternative to previous bonding techniques. Thus, for the specimen with the shortest free bonded length the results are even better than those obtained in series 4 (see the left-hand side of Fig. 20d) in terms of the cumulative energy and the global damage progression.

Lastly, this analysis suggests that an excellent CFRP-to-steel joint could be obtained in practice by combining the specific solutions of series 2, 4, and 5. This means that a stiffer and more ductile adhesive used on an extended bonded anchorage area (double layer and/or linear variable width), can ensure high levels of ultimate load and slip, of dissipated energy during cyclic loading, and sustain greater damage before failure.

5. Conclusions

In this work, different emerging bonding techniques applied to CFRP-to-steel joints subjected to monotonic and cyclic loading were numerically studied. The authors recognize that the present work would benefit from experimental validation. However, given the main goal of this work was to identify the best anchorage system for CFRP-to-steel joints, the numerical results obtained were sufficient to mitigate the doubts that all these emerging bonding techniques raise, prior to experimental testing. We can now follow up on the most promising solutions in the near future and add experimental data. Therefore, based on the results collected from all 48 simulations carried out and the subsequent analysis, the following main conclusions can be made:

- Since the interfacial slip at the failure of the CFRP-to-steel bonded joints increases with the overlapped bonded length, the ductility of the joints also increases. Subjected to a monotonic load, the use of a long bonded length has the advantage of keeping a meaningful bonded area undamaged that, in an unloading and reloading scenario, will allow the joint to preserve its initial strength;
- Based on the specimens subjected to the monotonic load, the highest load increase in the specimens with an additional anchorage system was achieved by linearly varying the width of the CFRP composite. Despite its smooth variation (between 10 mm and 30 mm along 150 mm length), the specimen with an anchorage length of 150 mm had an increase of 80.5% when compared to its homologous reference specimens, i.e. the EBR with the same total bonded length. Despite not being analysed here, it should be mentioned that the combination of this anchorage solution with one of the other solutions in Series 3 to 5 would have further improved the final strength of the specimens with a variable width of the CFRP composite;
- Unlike long anchorages, the use of an additional mechanical anchorage with a short length has almost no impact on the bond performance of the CFRP-to-steel joints, and, therefore, their use should be avoided since no benefit can be obtained from its use;
- The cyclic loading protocol applied to the specimens showed that the first of three cycles with the same interfacial slip was responsible for the highest load degradation of the CFRP-to-steel specimens. However, the tested specimens with the maximum total bonded length revealed that the joint maintained its original strength (in reference specimens under monotonic load), despite a significant number of cycles being applied (7 to 15 in 30);
- With the increase in the number of cycles of the cyclic loading protocol, the stiffness of the CFRP-to-steel joints decreases. The decrease in stiffness of the bonded joints is followed by the increase of the debonded length. Therefore, the global damage to the joints tends to increase with the number of cycles. Based on the results, the additional anchorage that was most influenced by the cyclic bond behaviour of the CFRP-to-steel joints was in the specimens with an additional anchorage of 150 mm in Series 5, i.e. with the addition of a steel plate bonded onto the CFRP composite. Compared with its homologous reference specimen, i.e. the one with the same total bonded length, an MRPDM approximately equal to 13.9% was calculated. Nevertheless, the highest calculated MRPDM of 77.9% was obtained from specimens with a short free bonded length perfectly anchored at the CFRP unpulled end (in Series 6). Therefore, the strength of CFRP-to-steel bonded joint is only maximized when using a bonding technique able to constrain the interfacial slips at the CFRP pulled end;

- During a cyclic loading scenario, two main aspects should be considered and used simultaneously to obtain a CFRP-to-steel bonded joint with a good ability to dissipate energy. One is to consider long free bonded length as well as long anchorage length. The other one is to use an additional anchorage like that simulated in Series 4, i.e. using an adhesive that permits the development of high interfacial maximum bond stress and, at the same time, has a meaningful softening stage. In contrast, the results obtained from the specimens that had a perfect anchorage registered the lowest dissipated energy values, and therefore, this anchorage system was identified as being the worst choice for dissipating energy in a cyclic loading scenario.

It should also be mentioned that despite being outside the scope of this study, the best bonding system to face cyclic loading could possibly be found in a combination of the bonding systems studied here. For instance, smoothly varying the width of the CFRP composite and, simultaneously, adopting another adhesive and/or installing a steel plate onto the surface of the CFRP composite, which will certainly provide interesting results to be pursued by the authors soon.

CRediT authorship contribution statement

Hugo Biscaia: Conceptualization, Validation, Investigation, Resources, Data curation, Formal analysis, Writing - original draft, Methodology, Writing - Review & Editing, Visualization, Supervision, Project administration, Funding acquisition. **Rui Micaelo:** Conceptualization, Validation, Investigation, Resources, Data curation, Formal analysis, Writing - original draft, Methodology, Writing - Review & Editing, Visualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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